CHOW AND CHAIN POLYNOMIALS OF MATROIDS ARE NOT LOG-CONCAVE

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Abstract. The Hilbert–Poincaré series of the Chow rings of the toric varieties of matroids, often called the *Chow polynomials*, and the f-polynomial of their boundary complexes, called the *chain polynomials*, have been conjectured to only have real zeros. We construct several explicit infinite families of matroids from finite geometries, whose multichain counts fail to be log-concave. They come from a common geometric construction called *hyperplane coalescence*, together with several classical operations in projective geometry and combinatorial designs. This induces failures of log-concavity for Chow, augmented Chow, chain, and h-polynomials of several explicit infinite families of matroids, disproving the real-rootedness conjectures of Ferroni–Schröter, Huh–Stevens, and Athanasiadis–Kalampogia-Evangelinou from the lowest possible ranks.

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1. Introduction

Let M be a loopless matroid of positive rank r, with lattice of flats $\mathscr{L} = \mathscr{L}(\mathrm{M})$, and let $\overline{\mathscr{L}} = \mathscr{L} \setminus \{\widehat{0}, \widehat{1}\}$. Feichtner and Yuzvinsky introduced the Chow ring $\underline{\mathrm{CH}}(\mathrm{M})$ as the Chow ring of the toric variety associated to the nested set fan for the maximal building set [FY04]. The boundary complex of this toric variety is the order complex $\Delta(\overline{\mathscr{L}})$ of $\overline{\mathscr{L}}$. The *Chow polynomial* and *chain polynomial* of M are, respectively, the Hilbert–Poincaré polynomial of $\underline{\mathrm{CH}}(\mathrm{M})$ and the face polynomial of this boundary complex:

$$\underline{\mathrm{H}}(\mathrm{M}; t) = \sum_{i=0}^{r-1} \dim \underline{\mathrm{CH}}^i(\mathrm{M}) t^i, \qquad f(\mathrm{M}; t) \coloneqq \sum_{i \geqslant 0} f_{i-1}(\Delta(\overline{\mathscr{L}})) t^i.$$

Later on in [BHM+22], Braden, Huh, Matherne, Proudfoot and Wang introduced the augmented toric variety of a matroid M, and defined the augmented Chow ring $\mathrm{CH}(\mathrm{M})$ as its Chow ring. Its Hilbert–Poincaré polynomial is the *augmented Chow*

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polynomial:

$$\mathrm{H}(\mathrm{M};t) = \sum_{i=0}^{r} \dim \mathrm{CH}^i(\mathrm{M}) t^i.$$

These polynomial invariants of matroids and of the underlying toric varieties have been proven to enjoy many positivity properties, coming from combinatorial Hodge theory, and topology of the boundary complex. In the seminal paper by Adiprasito, Huh and Katz [AHK18], the Chow ring of matroids, despite being the Chow ring of a possibly incomplete toric variety, satisfies the Kähler package, similar to the cohomology ring of a smooth projective variety. In particular, as a result of Poincaré duality and the hard Lefschetz theorem, the Hilbert–Poincaré series have symmetric, unimodal coefficients. Furthermore, Braden, Huh, Matherne, Proudfoot and Wang proved the corresponding statements for the augmented Chow ring [BHM$^+$22, Theorem 1.6].

For the chain polynomials, Coron, Ferroni and Li studied structural properties of the nested set complexes in the most general setting of arbitrary matroids with building sets [CFL26b]. They proved that reduced nested set complexes of matroids with arbitrary building sets are vertex decomposable, hence shellable, and admit convex ear decompositions [CFL26b, Corollary 1.2 and Theorem 1.3]. These imply, for the maximal building sets, that the h-polynomial of the order complex written as $h(\mathrm{M};t) = \sum_{i=0}^{d} h_i t^i$ with $d = r-1$ is strongly flawless:

$$h_0 \leqslant h_1 \leqslant \cdots \leqslant h_{\lfloor d/2 \rfloor}, \qquad h_i \leqslant h_{d-i} \quad (0 \leqslant i \leqslant \lfloor d/2 \rfloor).$$

In particular, it is top-heavy [CFL26b, Corollary 1.4]. Vertex decomposition, which implies shellability, also implies the nonnegativity of the h-vector and implies that the coefficient sequence $(f_{-1}(\overline{\mathscr{L}}), \ldots, f_{d-1}(\overline{\mathscr{L}}))$ of $f(\mathrm{M};t)$ is strongly flawless [Björ92, Proposition 7.2.5(i)].

However, stronger positivity properties of polynomials need to come beyond the Kähler package. For instance, Ferroni, Matherne, Stevens and Vecchi proved γ-positivity for both polynomials $\underline{\mathrm{H}}(\mathrm{M};t)$ and $\mathrm{H}(\mathrm{M},t)$ [FMSV24] as an immediate application of the semi-small decomposition of the Chow ring and the augmented Chow ring [BHM$^+$22].

Real-rootedness is widely considered as the strongest positivity property that a polynomial with nonnegative coefficients enjoys, and it is also invisible to the Hodge theory. For palindromic polynomials, it simultaneously implies γ-positivity, log-concavity, and unimodality. Along this line, Huh and Stevens, and independently Ferroni and Schröter, proposed the following conjecture for the Chow polynomials of matroids.

Conjecture 1.1 ([Ste21, Conjecture 4.1.3] [FS24, Conjecture 8.18]) For any loopless matroid M, the polynomial $\underline{\mathrm{H}}(\mathrm{M};t)$ is real-rooted.

Huh and Stevens proposed the augmented analogue [Ste21, Conjecture 4.3.3]; see also [FMSV24, Conjecture 1.6].

Conjecture 1.2 (Huh–Stevens) For any loopless matroid M, the polynomial $\mathrm{H}(\mathrm{M};t)$ is real-rooted.

Around the same time, Athanasiadis and Kalampogia-Evangelinou conjectured the following analogous statement [AKE23, Conjecture 1.2].

Conjecture 1.3 (Athanasiadis–Kalampogia-Evangelinou) For any loopless matroid M, the polynomial $f(\mathrm{M};t)$ is real-rooted.

These conjectures have stimulated much research activity in recent years. Several classes of matroids satisfy these conjectures. Athanasiadis and Kalampogia-Evangelinou proved real-rootedness of chain polynomials for subspace lattices, partition lattices and their type B analogues, and recorded the uniform and near-pencil cases [AKE23, Theorem 1.3]. Brändén and Leite established it for perfect matroid designs, Dowling geometries and paving matroids [BL25], and for all geometric lattices of rank at most 3 [BL25, Theorem 6.1].

For Chow polynomials, Brändén and Vecchi proved real-rootedness for uniform matroids [BV25b, Theorem 1.1]. The augmented uniform case was proved by Ferroni, Matherne, Stevens and Vecchi [FMSV24, Theorem 1.10]. Hoster and Stump gave a common extension to finite graded simplicial posets with nonnegative h-vectors, after adjoining a top element [HS25, Theorem 1.1]. Brändén and Vecchi also proved real-rootedness of Chow and augmented Chow polynomials for totally nonnegative posets and their duals, and for paving matroids [BV25a, Theorem 6.1 and Corollary 7.2].

Concurrently, Coron, Ferroni and Li introduced UMEL-shellable posets, which are posets admitting an EL-labeling satisfying uniformity and monotonicity conditions. They proved real-rootedness of these polynomials for UMEL-shellable posets [CFL25, Theorems 1.2 and 1.5]. For matroids, UMEL-shellable posets include prominent families of matroids such as uniform matroids, affine and projective geometries, braid matroids of types A and B, all Dowling geometries, and their truncations and rank-selections [CFL25, Theorem 1.6, Theorem 1.8 and Proposition 7.1]. In recent work on matroid analogues of Gal's conjecture, Coron, Ferroni and Li further proved that $\underline{\mathrm{H}}(\mathrm{M};t)$ is real-rooted for every matroid of rank at most 6 [CFL26a, Corollary 8.5].

The main goal of this paper is to construct explicit infinite families of counterexamples to all the conjectures above. They share a common source which comes from the failure of log-concavity of multichain counts of matroids. The *multichain counts* of a loopless matroid M are defined by the function $\mathbb{Z}_{\geqslant 0} \to \mathbb{Z}_{\geqslant 0}$

$$Z_{\mathrm{M}}(k) \coloneqq \#\{\widehat{0} = F_0 \leqslant F_1 \leqslant \cdots \leqslant F_k = \widehat{1} \in \mathscr{L}(\mathrm{M})\},$$

which counts the multichains of $\mathscr{L}(\mathrm{M})$ by size. See [Sta12, Section 3.12]. We have the following amplification principle.

Theorem A *Let* M *be a loopless matroid. If there is an integer* $j \geqslant 2$ *such that*

$$Z_{\mathrm{M}}(j)^2 < Z_{\mathrm{M}}(j-1)Z_{\mathrm{M}}(j+1),$$

then for every sufficiently large integer n*, the polynomials*

$$f(\mathrm{M}^{\oplus n};t), \qquad h(\mathrm{M}^{\oplus n};t), \qquad \underline{\mathrm{H}}(\mathrm{M}^{\oplus n};t), \qquad \mathrm{H}(\mathrm{M}^{\oplus n};t)$$

are not log-concave, and hence not real-rooted.

We construct four families of matroids from finite geometries whose multichain counts satisfy the hypothesis of Theorem A. All four families use a geometric technique which we call *hyperplane coalescence*. In a nutshell, hyperplane coalescence merges selected collections of hyperplanes into larger hyperplanes while preserving every flat of lower rank, enabling us to alter chain counts without changing the lower-rank geometry of the matroid. Hyperplane coalescence generalizes the classical operation of tightening a free basis, which is the inverse of *circuit-hyperplane relaxation*, and can

be viewed as a matroid operation that naturally arises in the Crapo–Knuth theory of detruncations [Cra70, Knu75]; see also [PvdP17, Section 3].[1]

(1) The *affine Klein family* M_q of rank 4, for odd prime powers q, constructed from affine geometry using spreads obtained from the Klein correspondence [Hir85].
(2) The *affine packing family* $\mathrm{M}_q^{\mathrm{pk}}$ of rank 4, for prime powers q, constructed from affine geometry using Denniston's packings of projective lines into spreads [Den72, Den73].
(3) The *Gabidulin family* $\mathrm{G}_{k,q}$ of rank k, for $k \geqslant 4$ and prime powers q, constructed from projective geometry using rank-metric codes [Del78, Gab85].
(4) The *projective family* $\mathrm{C}(q, N)$ of rank 7, for prime powers q and integers $4 \leqslant N \leqslant q^3$, constructed from projective planes and auxiliary four-element sets using rank-distance bounds [Del78, Gab85].

The parameter range depends on the family: M_q requires an odd prime power, whereas $\mathrm{M}_q^{\mathrm{pk}}$, $\mathrm{G}_{k,q}$, and $\mathrm{C}(q, N)$ allow any prime power, including $q = 2^a$ for $a \geqslant 1$.

Theorem B *The following equivalences hold:*

(a) For every odd prime power q,

$$Z_{\mathrm{M}_q}(3)^2 < Z_{\mathrm{M}_q}(2)Z_{\mathrm{M}_q}(4) \quad \Longleftrightarrow \quad q \geqslant 97.$$

(b) For every prime power q,

$$Z_{\mathrm{M}_q^{\mathrm{pk}}}(3)^2 < Z_{\mathrm{M}_q^{\mathrm{pk}}}(2)Z_{\mathrm{M}_q^{\mathrm{pk}}}(4) \quad \Longleftrightarrow \quad q \geqslant 81.$$

(c) For every prime power q,

$$Z_{\mathrm{G}_{4,q}}(3)^2 < Z_{\mathrm{G}_{4,q}}(2)Z_{\mathrm{G}_{4,q}}(4) \quad \Longleftrightarrow \quad q \geqslant 81.$$

(d) For every prime power q,

$$Z_{\mathrm{C}(q,q^3)}(3)^2 < Z_{\mathrm{C}(q,q^3)}(2)Z_{\mathrm{C}(q,q^3)}(4) \quad \Longleftrightarrow \quad q \geqslant 149.$$

For each matroid M satisfying one of these inequalities and every sufficiently large integer n, the polynomials

$$f(\mathrm{M}^{\oplus n}; t), \qquad h(\mathrm{M}^{\oplus n}; t), \qquad \underline{\mathrm{H}}(\mathrm{M}^{\oplus n}; t), \qquad \mathrm{H}(\mathrm{M}^{\oplus n}; t)$$

are neither log-concave nor real-rooted.

Example 1.4 The matroid M_{97} is on 88 533 986 elements of rank 4, with

$$\begin{aligned} Z_{\mathrm{M}_{97}}(2) &= 981\,796\,552\,261\,343, \\ Z_{\mathrm{M}_{97}}(3) &= 28\,747\,469\,795\,558\,806, \\ Z_{\mathrm{M}_{97}}(4) &= 898\,967\,215\,589\,248\,050. \end{aligned}$$

These numbers of multichains satisfy

$$\frac{Z_{\mathrm{M}_{97}}(3)^2}{Z_{\mathrm{M}_{97}}(2)Z_{\mathrm{M}_{97}}(4)} \approx 0.93634 < 1.$$

[1] Throughout this paper, we use *detruncation* for what is classically called a nontrivial *erection*: Given a matroid M, a *detruncation* of M is a rank-r matroid N whose rank-$(r-1)$ truncation is M. Our terminology emphasizes its relation to truncation.

The chain polynomial of the full lattice $\mathscr{L}(\mathrm{M}_{97})$ is

$$\begin{aligned} f(\mathscr{L}(\mathrm{M}_{97});t) = \ & 1 + 981\,796\,552\,261\,343t + 27\,765\,673\,243\,297\,463t^2 \\ & + 842\,454\,072\,550\,391\,781t^3 + 1\,605\,538\,311\,579\,936\,540t^4 \\ & + 789\,868\,115\,720\,580\,880t^5, \end{aligned}$$

with the coefficients a_1, a_2, a_3 of the terms t, t^2, t^3 satisfying

$$a_2^2 - a_1 a_3 = -56\,185\,893\,214\,937\,313\,548\,622\,648\,985\,514 < 0.$$

The matroid $\mathrm{N} = \mathrm{M}_{97} \oplus \mathrm{M}_{97}$ has rank 8 on 177 067 972 elements with Chow polynomial

$$\begin{aligned} \underline{\mathrm{H}}(\mathrm{N};t) = 1 & + 963\,924\,470\,032\,260\,016\,646\,795\,095\,676t \\ & + 772\,700\,986\,375\,783\,620\,699\,737\,483\,420\,137t^2 \\ & + 668\,902\,642\,263\,664\,767\,416\,856\,175\,849\,115\,238t^3 \\ & + 668\,902\,642\,263\,664\,767\,416\,856\,175\,849\,115\,238t^4 \\ & + 772\,700\,986\,375\,783\,620\,699\,737\,483\,420\,137t^5 \\ & + 963\,924\,470\,032\,260\,016\,646\,795\,095\,676t^6 + t^7. \end{aligned}$$

The coefficients a_1, a_2, a_3 of the terms t, t^2, t^3 satisfy

$$\frac{a_2^2}{a_1 a_3} \approx 0.926013 < 1.$$

The projective family gives Chow counterexamples in rank 7. For every prime power $q \geqslant 512$, the matroid $\mathrm{C}(q, q^3)$ is simple and 3-connected, and its Chow polynomial is not log-concave, and hence not real-rooted. More generally, the same conclusions hold for $\mathrm{C}(q, \lfloor \rho\, q^3 \rfloor)$ for each fixed $0 < \rho \leqslant 1$ and all sufficiently large prime powers q.

Example 1.5 The matroid $\mathrm{C} = \mathrm{C}(512, 512^3)$ is simple and 3-connected, of rank 7, on

$$1 + 512 + 512^2 + 2 \cdot 512^3 + 512^4 + 512^5 = 35\,253\,360\,263\,681$$

elements. Its Chow polynomial is

$$\begin{aligned} \underline{\mathrm{H}}(\mathrm{C};t) = 1 & + 127962875286940823606943757091077833817717339835617 34t \\ & + 226086791088516634983818986911539069056896541442363393 5t^2 \\ & + 402548512894232793390837832528792797279036205852996415508t^3 \\ & + 226086791088516634983818986911539069056896541442363393 5t^4 \\ & + 127962875286940823606943757091077833817717339835617 34t^5 + t^6. \end{aligned}$$

The coefficients a_1, a_2, a_3 of the terms t, t^2, t^3 satisfy

$$\frac{a_2^2}{a_1 a_3} \approx 0.992312 < 1.$$

Thus $\underline{\mathrm{H}}(\mathrm{C};t)$ is not log-concave, and hence not real-rooted.

Remark 1.6 These families achieve the smallest possible ranks for failures of both log-concavity and real-rootedness of the h-polynomial and the Chow polynomial.

Brändén and Leite proved that the chain polynomial and the h-polynomial of every geometric lattice of rank at most 3 are real-rooted [BL25, Theorem 6.1]. For Chow polynomials, Coron, Ferroni and Li proved real-rootedness for every matroid of rank at most 6 [CFL26a, Corollary 8.5].

The nontrivial direct powers in Theorem A are disconnected and have large ground sets. Below, we obtain rank-7 counterexamples to the log-concavity of Chow polynomials directly from the rank-4 counterexamples to the log-concavity of the chain polynomials, by truncating the direct sum of two members in the latter family or taking their parallel connections. These constructions give counterexamples that are minimal in their ranks and have much smaller ground sets. These examples also distinguish failure of real-rootedness from failure of log-concavity. For example, $\mathrm{Tr}^7(\mathrm{M}_q \oplus \mathrm{M}_q)$ has a log-concave but non-real-rooted Chow polynomial precisely for odd prime powers $47 \leqslant q \leqslant 89$, and its Chow polynomial is not log-concave for $q \geqslant 97$. Adding one coloop before truncation gives 3-connected examples with the same thresholds (Theorem C).

Theorem C *The matroids*

$$\mathrm{Tr}^7(\mathrm{M}_q \oplus \mathrm{M}_q \oplus \mathrm{U}_{1,1}), \qquad \mathrm{Tr}^7(\mathrm{G}_{4,q} \oplus \mathrm{G}_{4,q} \oplus \mathrm{U}_{1,1})$$

are simple, 3-connected and of rank 7. The following hold:

(a) For every odd prime power q, $\underline{\mathrm{H}}(\mathrm{Tr}^7(\mathrm{M}_q \oplus \mathrm{M}_q \oplus \mathrm{U}_{1,1}); t)$ is not real-rooted exactly when $q \geqslant 47$, and is not log-concave exactly when $q \geqslant 97$.

(b) For every prime power q, $\underline{\mathrm{H}}(\mathrm{Tr}^7(\mathrm{G}_{4,q} \oplus \mathrm{G}_{4,q} \oplus \mathrm{U}_{1,1}); t)$ is not real-rooted exactly when $q \geqslant 37$, and is not log-concave exactly when $q \geqslant 81$.

Corollary D *Conjecture 1.1, Conjecture 1.2 and Conjecture 1.3 are false.*

Proof. The inequalities for the numbers of multichains in Theorem B, together with Theorem A, give non-log-concavity of the chain, Chow, and augmented Chow polynomials. By Newton's inequalities, the three real-rootedness conjectures are false. □

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Organization. Section 2 reviews the polynomial invariants and proves Theorem A. Section 3 develops hyperplane coalescence and absorption, and studies the changes of the multichain counts under coalescence. Section 4 constructs the four geometric families, proves Theorem B in Section 4.4, and derives the 3-connected rank-7 counterexamples of Theorem C in Section 4.5.

2. Polynomial invariants and multichain counts

In this section, we review basic properties of the relevant polynomial invariants of matroids, and prove Theorem A. We then develop the parallel amplification argument for graph associahedra under wedges. Throughout, M is a loopless matroid of positive rank r, $\mathscr{L}$ denotes the lattice of flats of M, and $\overline{\mathscr{L}}$ denotes the poset of nonempty proper flats of M.

2.1. **Chain and face enumeration polynomials.** Let $f_{k-1}(\mathscr{L})$ be the number of faces of dimension $k-1$ in the order complex of $\mathscr{L}$, with $f_{-1}=1$ for the empty face. The face polynomials of $\mathscr{L}$ and $\overline{\mathscr{L}}$ are defined by

$$f(\mathscr{L};t)=\sum_{k\geqslant 0} f_{k-1}(\mathscr{L})t^k, \qquad f(\overline{\mathscr{L}};t)=\sum_{k=0}^{r-1} f_{k-1}(\overline{\mathscr{L}})t^k.$$

For a matroid M, write $f(\mathrm{M};t)=f(\overline{\mathscr{L}};t)$. For $S\subseteq[r-1]=\{1,\dots,r-1\}$, let $\alpha_{\mathrm{M}}(S)$ count the chains in $\overline{\mathscr{L}}$ with rank set S. These numbers form the *flag f-vector*. The *flag h-vector* is defined by

$$\beta_{\mathrm{M}}(S)=\sum_{T\subseteq S}(-1)^{|S\setminus T|}\alpha_{\mathrm{M}}(T).$$

In particular, $W_{\mathrm{M}}(i)=\alpha_{\mathrm{M}}(i)$ is the number of rank-i flats, and $\alpha_{\mathrm{M}}(\varnothing)=\beta_{\mathrm{M}}(\varnothing)=1$.

The order complex of $\overline{\mathscr{L}}$ has dimension $r-2$ and face numbers

$$f_{k-1}(\overline{\mathscr{L}})=\sum_{\substack{S\subseteq[r-1]\\|S|=k}}\alpha_{\mathrm{M}}(S).$$

The h-polynomial of $\overline{\mathscr{L}}$ is defined by

$$h(\mathrm{M};t)\coloneqq\sum_{k=0}^{r-1} f_{k-1}(\overline{\mathscr{L}})t^k(1-t)^{r-1-k}=\sum_{S\subseteq[r-1]}\beta_{\mathrm{M}}(S)t^{|S|}. \tag{1}$$

The flag h-numbers are nonnegative, and $\beta_{\mathscr{L}}([r-1])=|\mu_{\mathscr{L}}(\widehat{0},\widehat{1})|$ [Bjö92]. The chain and h-polynomials are related by

$$f(\mathscr{L};t)=(1+t)^2 f(\overline{\mathscr{L}};t), \qquad f(\overline{\mathscr{L}};t)=(1+t)^{r-1}h\left(\mathrm{M};\frac{t}{1+t}\right). \tag{2}$$

2.2. **Multichain counts.** We recall the background of the multichain counting polynomial introduced in [Sta12, Section 3.12] in the setting of matroids. For a loopless matroid M of rank r, the function Z_{M} counts multichains in its lattice of flats:

$$Z_{\mathrm{M}}(m)\coloneqq\#\{\widehat{0}=F_0\leqslant F_1\leqslant\cdots\leqslant F_m=\widehat{1}\} \qquad (m\geqslant 0).$$

In particular, $Z_{\mathrm{M}}(0)=0$ and $Z_{\mathrm{M}}(1)=1$. For $m\geqslant 2$, a multichain counted by $Z_{\mathrm{M}}(m)$ is determined by its $m-1$ intermediate flats in $\mathscr{L}$, which may include $\widehat{0}$ and $\widehat{1}$. If

these entries contain k distinct flats, they form a strict chain counted by $f_{k-1}(\mathscr{L})$. Their positive multiplicities sum to $m-1$, giving $\binom{m-2}{k-1}$ choices. Therefore,

$$Z_{\mathrm{M}}(m) = \sum_{k=1}^{r+1} f_{k-1}(\mathscr{L})\binom{m-2}{k-1} = \sum_{k=0}^{r-1} f_{k-1}(\overline{\mathscr{L}})\binom{m}{k+1} \qquad (m \geqslant 2), \tag{3}$$

with $Z_{\mathrm{M}}(0) = 0$, $Z_{\mathrm{M}}(1) = 1$, and $Z_{\mathrm{M}}(m)$ is strictly increasing for integers $m \geqslant 1$. In particular,

$$\begin{aligned} f_0(\mathscr{L}) &= Z_{\mathrm{M}}(2), \\ f_1(\mathscr{L}) &= Z_{\mathrm{M}}(3) - Z_{\mathrm{M}}(2), \\ f_2(\mathscr{L}) &= Z_{\mathrm{M}}(4) - 2Z_{\mathrm{M}}(3) + Z_{\mathrm{M}}(2). \end{aligned}$$

Lemma 2.1 *For any matroid* M*,*

$$f_1(\mathscr{L})^2 - f_0(\mathscr{L})f_2(\mathscr{L}) = Z_{\mathrm{M}}(3)^2 - Z_{\mathrm{M}}(2)Z_{\mathrm{M}}(4). \tag{4}$$

Thus $f(\mathscr{L};t)$ fails log-concavity at coefficient index 2 exactly when the values of the zeta polynomial fail log-concavity at $m = 3$.

One advantage of the zeta polynomial is its behavior over direct sums: a multichain in a product lattice is a tuple of multichains in its factors. By definition of multichain counts, we have

$$Z_{\mathrm{M}^{\oplus n}}(m) = Z_{\mathrm{M}}(m)^n. \tag{5}$$

Lemma 2.2 *If for $j \geqslant 2$,*

$$Z_{\mathrm{M}}(j)^2 < Z_{\mathrm{M}}(j-1)Z_{\mathrm{M}}(j+1)$$

then the same inequality holds for $\mathrm{M}^{\oplus n}$ for every $n \geqslant 1$. Furthermore, if $j = 3$, then $f(\mathscr{L}(\mathrm{M}^{\oplus n});t)$ fails to be log-concave at coefficient index 2.

Proof. The first statement follows from (5):

$$\frac{Z_{\mathrm{M}^{\oplus n}}(j)^2}{Z_{\mathrm{M}^{\oplus n}}(j-1)Z_{\mathrm{M}^{\oplus n}}(j+1)} = \left(\frac{Z_{\mathrm{M}}(j)^2}{Z_{\mathrm{M}}(j-1)Z_{\mathrm{M}}(j+1)}\right)^n < 1.$$

The second statement follows from (4). □

2.3. **Chow polynomials and augmented Chow polynomials.** The Chow ring $\underline{\mathrm{CH}}(\mathrm{M})$ and the augmented Chow ring $\mathrm{CH}(\mathrm{M})$ have Hilbert–Poincaré polynomials

$$\underline{\mathrm{H}}(\mathrm{M};t) = \sum_{i=0}^{r-1} \dim \underline{\mathrm{CH}}^i(\mathrm{M})t^i, \quad \mathrm{H}(\mathrm{M};t) = \sum_{i=0}^{r} \dim \mathrm{CH}^i(\mathrm{M})t^i.$$

We first describe the Chow polynomial using chains of flats. For a nonempty subset $S = \{s_1 < \cdots < s_k\} \neq \varnothing$, let

$$\begin{aligned} \underline{\tau}(S) &= t[s_1 - 1]_t \prod_{i=2}^{k} t[s_i - s_{i-1} - 1]_t, \\ \tau(S) &= t[s_1]_t \prod_{i=2}^{k} t[s_i - s_{i-1} - 1]_t, \end{aligned}$$

with $\underline{\tau}(\varnothing) = \tau(\varnothing) = 1$. Grouping chains by their proper-flat rank sets gives

$$\begin{aligned} \underline{\mathrm{H}}(\mathrm{M};t) &= \sum_{S\subseteq[r-1]} \alpha_{\mathrm{M}}(S)\big(\underline{\tau}(S) + \underline{\tau}(S\cup\{r\})\big), \\ \mathrm{H}(\mathrm{M};t) &= \sum_{S\subseteq[r-1]} \alpha_{\mathrm{M}}(S)\big(\tau(S) + \tau(S\cup\{r\})\big). \end{aligned} \tag{6}$$

The Chow and augmented Chow polynomials can be expressed using the flag h-vectors [Stu24, Theorems 1.1 and 1.2]. A set $S \subseteq [r-1]$ is *stable* if it contains no consecutive integers. Then

$$\begin{aligned} \underline{\mathrm{H}}(\mathrm{M};t) &= \sum_{\substack{S\subseteq[r-1] \text{ stable} \\ 1\notin S}} \beta_{\mathrm{M}}(S) t^{|S|}(1+t)^{r-1-2|S|}, \\ \mathrm{H}(\mathrm{M};t) &= \sum_{S\subseteq[r-1] \text{ stable}} \beta_{\mathrm{M}}(S) t^{|S|}(1+t)^{r-2|S|}. \end{aligned}$$

2.4. **Asymptotic coefficients.** In this section, we give asymptotic formulas for the coefficients of the aforementioned polynomial invariants of the direct sum $\mathrm{M}^{\oplus n}$ in terms of the multichain counts of M. First, we have the following explicit formula for every coefficient of the face enumeration polynomials of $\mathrm{M}^{\oplus n}$.

Proposition 2.3 *For $n \geqslant 1$ and $j \geqslant 0$,*

$$\begin{aligned} f_{j-1}(\mathrm{M}^{\oplus n}) &= \sum_{m=1}^{j+1} (-1)^{j+1-m} \binom{j+1}{m} Z_{\mathrm{M}}(m)^n, \\ h_j(\mathrm{M}^{\oplus n}) &= \sum_{m=1}^{j+1} (-1)^{j+1-m} \binom{nr+1}{j+1-m} Z_{\mathrm{M}}(m)^n. \end{aligned}$$

Proof. The first identity follows by binomial inversion from Equation 3 and Equation 5. For the second identity, the identity

$$\sum_{m\geqslant 0} \binom{m}{k+1} t^m = \frac{t^{k+1}}{(1-t)^{k+2}}$$

implies that

$$\sum_{m\geqslant 1} Z_{\mathrm{M}^{\oplus n}}(m) t^{m-1} = \sum_{k=0}^{nr-1} f_{k-1}(\mathrm{M}^{\oplus n}) \frac{t^k}{(1-t)^{k+2}} = \frac{h(\mathrm{M}^{\oplus n};t)}{(1-t)^{nr+1}}.$$

Multiply by $(1-t)^{nr+1}$, substitute $Z_{\mathrm{M}^{\oplus n}}(m) = Z_{\mathrm{M}}(m)^n$, and compare coefficients. □

For the asymptotic coefficients of the Chow and augmented Chow polynomials of $\mathrm{M}^{\oplus n}$, we express the polynomials using incidence-algebraic techniques from the work of [FMV24]. The incidence algebra $\mathcal{I}(\mathscr{L})$ is an associative algebra, consisting of polynomials on closed intervals of $\mathscr{L}$. For a more general description of incidence algebra for posets, see [Sta12, Section 3.7]. In $\mathcal{I}(\mathscr{L})$, multiplication is convolution

$$(a\cdot b)(x,y;t) \coloneqq \sum_{x\leqslant u\leqslant y} a(x,u;t)\, b(u,y;t), \text{ for every } x\leqslant y \text{ in } \mathscr{L},$$

and the multiplicative identity of $\mathcal{I}(\mathcal{L})$ is the delta function

$$\delta(x,y) \coloneqq \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases}.$$

We also have the zeta function, together with the Möbius function as its inverse:

$$\zeta(x,y) \coloneqq \begin{cases} 1 & \text{if } x \leqslant y \\ 0 & \text{otherwise} \end{cases}, \qquad \mu(x,y) \coloneqq \begin{cases} 1 & \text{if } x = y \\ -\sum_{x \leqslant u < y} \mu(x,u) & \text{if } x < y \end{cases}.$$

For $\mathcal{L}$, $\mathcal{I}(\mathcal{L})$ admits an involution $a \mapsto a^{\mathrm{rev}}$ defined by

$$a^{\mathrm{rev}}(x,y;t) \coloneqq t^{\mathrm{rk_M}(y)-\mathrm{rk_M}(x)} a(x,y;t^{-1}),$$

which reverses the coefficients of the polynomials $a(x,y;t)$ on an interval $[x,y]$. The *characteristic function* is defined as

$$\chi \coloneqq \mu \cdot \zeta^{\mathrm{rev}}, \text{ with } \zeta^{\mathrm{rev}}(x,y;t) = t^{\mathrm{rk}(y)-\mathrm{rk}(x)},$$

and for $\mathcal{L}$, $\chi(\widehat{0},\widehat{1})$ coincides with the characteristic polynomial of the matroid M. Following the construction and result in [FMV24, Definitions 3.2, 3.3, 3.13 and Theorems 4.9, 4.11], the Chow and left augmented Chow functions are defined as

$$\begin{aligned} \underline{\mathrm{H}} &\coloneqq (1-t)(\chi - t\delta)^{-1} = (1-t)\sum_{m \geqslant 1} t^{m-1}\chi^{-m}, \\ \mathrm{H} &\coloneqq \zeta^{\mathrm{rev}} \cdot \underline{\mathrm{H}} = (1-t)\sum_{m \geqslant 1} t^{m-1}\big(\zeta^{\mathrm{rev}} \cdot \chi^{-m}\big), \end{aligned} \tag{7}$$

with $\underline{\mathrm{H}}(\widehat{0},\widehat{1};t) = \underline{\mathrm{H}}(\mathrm{M};t)$ and $\mathrm{H}(\widehat{0},\widehat{1};t) = \mathrm{H}(\mathrm{M};t)$.

Proposition 2.4 *For every $n \geqslant 1$,*

$$\begin{aligned} \underline{\mathrm{H}}(\mathrm{M}^{\oplus n};t) &= (1-t)\sum_{m \geqslant 1} t^{m-1}\big((\chi^{-m})(\widehat{0},\widehat{1};t)\big)^n, \\ \mathrm{H}(\mathrm{M}^{\oplus n};t) &= (1-t)\sum_{m \geqslant 1} t^{m-1}\big((\zeta^{\mathrm{rev}} \cdot \chi^{-m})(\widehat{0},\widehat{1};t)\big)^n. \end{aligned} \tag{8}$$

Proof. Since $\mathcal{L}(\mathrm{M}^{\oplus n}) \cong \mathcal{L}(\mathrm{M})^n$, the functions ζ, ζ^{rev}, and χ in each summand of Equation 7 factor over the direct sum. Convolution and taking inverses respect this factorization, and the formulas then follow. □

We have the following computational lemma relating the expressions above to multichain counts.

Lemma 2.5 *For any fixed $m \geqslant 1$, and a polynomial $Q_m(\mathrm{M};t)$ equal to*

$$(\chi_{\mathrm{M}}^{-m})(\widehat{0},\widehat{1};t) \quad \text{or} \quad (\zeta_{\mathrm{M}}^{\mathrm{rev}} \cdot \chi_{\mathrm{M}}^{-m})(\widehat{0},\widehat{1};t),$$

we have the constant terms

$$Q_m(\mathrm{M};0) = \zeta_{\mathrm{M}}^m(\widehat{0},\widehat{1}) = Z_{\mathrm{M}}(m), \qquad Q_m(\mathrm{M}^{\oplus n};0) = Z_{\mathrm{M}}(m)^n.$$

Proof. We first consider the case of a single matroid M. For $(\chi_{\mathrm{M}}^{-m})(\widehat{0},\widehat{1};t)$, evaluation at $t = 0$ commutes with convolution and therefore also with inversion. Since $\chi_{\mathrm{M}}(0) = \mu_{\mathrm{M}}$ and $\mu_{\mathrm{M}}^{-1} = \zeta_{\mathrm{M}}$, we obtain

$$\chi_{\mathrm{M}}^{-m}(0) = \mu_{\mathrm{M}}^{-m} = \zeta_{\mathrm{M}}^m.$$

For $(\zeta_{\mathrm{M}}^{\mathrm{rev}} \cdot \chi_{\mathrm{M}}^{-m})(\widehat{0}, \widehat{1}; t)$, since $\zeta_{\mathrm{M}}^{\mathrm{rev}}(0) = \delta_{\mathrm{M}}$, which is the multiplicative identity, the constant term is unchanged:

$$(\zeta_{\mathrm{M}}^{\mathrm{rev}} \cdot \chi_{\mathrm{M}}^{-m})(0) = \chi_{\mathrm{M}}^{-m}(0) = \zeta_{\mathrm{M}}^{m}.$$

Therefore, $Q_m(\mathrm{M}; 0)$ is equal to

$$\zeta_{\mathrm{M}}^{m}(\widehat{0}, \widehat{1}) = \sum_{\widehat{0}=F_0 \leqslant F_1 \leqslant \cdots \leqslant F_m = \widehat{1}} \Big(\prod_{i=1}^{m} \zeta_{\mathrm{M}}(F_{i-1}, F_i) \Big) = \sum_{\widehat{0}=F_0 \leqslant F_1 \leqslant \cdots \leqslant F_m = \widehat{1}} 1 = Z_{\mathrm{M}}(m).$$

We next apply the same argument to $\mathrm{M}^{\oplus n}$. Since $\mathscr{L}(\mathrm{M}^{\oplus n}) \cong \mathscr{L}(\mathrm{M})^n$, a multichain in this product is determined by an independent choice of a multichain in each coordinate. Therefore,

$$Q_m(\mathrm{M}^{\oplus n}; 0) = Z_{\mathrm{M}^{\oplus n}}(m) = Z_{\mathrm{M}}(m)^n.$$

□

Proposition 2.6 *Let $P(\mathrm{M}; t)$ denote any one of*

$$f(\mathscr{L}; t), \qquad f(\overline{\mathscr{L}}; t), \qquad h(\mathrm{M}; t), \qquad \underline{\mathrm{H}}(\mathrm{M}; t), \qquad \mathrm{H}(\mathrm{M}; t),$$

and write $P(\mathrm{M}; t) = \sum_j P_j(\mathrm{M}) t^j$. For a fixed $j \geqslant 1$,

$$\lim_{n \to \infty} P_j(\mathrm{M}^{\oplus n}) = Z_{\mathrm{M}}(j+1)^n + O\big(n^j Z_{\mathrm{M}}(j)^n\big). \tag{9}$$

Furthermore,

$$\lim_{n \to \infty} \frac{P_j(\mathrm{M}^{\oplus n})}{Z_{\mathrm{M}}(j+1)^n} = 1.$$

Proof. For $f(\mathrm{M}; t)$ or $h(\mathrm{M}; t)$, since $Z_{\mathrm{M}}(m)$ is strictly increasing, Proposition 2.3 implies that as n goes to ∞, the jth coefficient of $f(\mathrm{M}^{\oplus n}; t)$, which is $f_{j-1}(\mathrm{M}^{\oplus n})$, or that of $h(\mathrm{M}^{\oplus n}; t)$, which is $h_j(\mathrm{M}^{\oplus n})$, is dominated by the term $Z_{\mathrm{M}}(j+1)^n$. Therefore, we have the asymptotic ratios

$$\lim_{n \to \infty} \frac{f_{j-1}(\mathrm{M}^{\oplus n})}{Z_{\mathrm{M}}(j+1)^n} = 1, \qquad \lim_{n \to \infty} \frac{h_j(\mathrm{M}^{\oplus n})}{Z_{\mathrm{M}}(j+1)^n} = 1.$$

By (2), the same limit is true for the jth coefficient of $f(\mathscr{L}; t)$.

For $\underline{\mathrm{H}}(\mathrm{M}; t)$ and $\mathrm{H}(\mathrm{M}; t)$, if we write

$$Q_m(\mathrm{M}^{\oplus n}; t) = \begin{cases} (\chi^{-m})(\widehat{0}, \widehat{1}; t) & \text{for } \underline{\mathrm{H}} \\ (\zeta^{\mathrm{rev}} \cdot \chi^{-m})(\widehat{0}, \widehat{1}; t) & \text{for } \mathrm{H} \end{cases},$$

then Proposition 2.4 implies that

$$P(\mathrm{M}^{\oplus n}; t) = \sum_{m \geqslant 1} (t^{m-1} - t^m) Q_m(\mathrm{M}^{\oplus n}; t).$$

Therefore, for $j \geqslant 1$, the jth coefficient of $P(\mathrm{M}^{\oplus n}; t)$ is

$$P_j(\mathrm{M}^{\oplus n}) = Q_{j+1}(\mathrm{M}; 0)^n + \sum_{m=1}^{j} \big(Q_{m,j+1-m}(\mathrm{M}^{\oplus n}) - Q_{m,j-m}(\mathrm{M}^{\oplus n}) \big).$$

Then Lemma 2.5 and Lemma 2.7 imply that, as n goes to infinity, the first term $Q_{j+1}(\mathrm{M}; 0)^n = Z_{\mathrm{M}}(j+1)^n$ dominates, and

$$\lim_{n \to \infty} \frac{P_j(\mathrm{M}^{\oplus n})}{Z_{\mathrm{M}}(j+1)^n} = 1.$$

□

Lemma 2.7 *For either choice of Q_m, and for every fixed $j \geqslant 1$, there exists a constant $C = C_{\mathrm{M},j}$ such that, as $n \to \infty$,*

$$\sum_{m=1}^{j} \big(Q_{m,j+1-m}(\mathrm{M}^{\oplus n}) - Q_{m,j-m}(\mathrm{M}^{\oplus n})\big) \leqslant C_{\mathrm{M},j} n^j Z_{\mathrm{M}}(j)^n.$$

Proof. By multiplicativity and Lemma 2.5,

$$Q_m(\mathrm{M}^{\oplus n}; t) = Q_m(\mathrm{M}; t)^n, \qquad c_m := Q_m(\mathrm{M}; 0) = Z_{\mathrm{M}}(m) > 0.$$

Write $Q_m(\mathrm{M}; t) = c_m(1 + tB_m(t))$, where $B_m(t)$ is independent of n. The binomial theorem implies

$$Q_{m,d}(\mathrm{M}^{\oplus n}) = c_m^n \sum_{\ell=0}^{\min(n,d)} \binom{n}{\ell} B_{m,d-\ell}(t)^\ell,$$

where the coefficients $B_{m,d-\ell}(t)^\ell$ are independent of n. Since $0 \leqslant \ell \leqslant \min(n, d)$,

$$\binom{n}{\ell} \leqslant \frac{n^\ell}{\ell!} \leqslant \frac{n^d}{\ell!},$$

and there exists a constant $C_{\mathrm{M},m,d}$ such that

$$|Q_{m,d}(\mathrm{M}^{\oplus n})| \leqslant C_{\mathrm{M},m,d}\, n^d Z_{\mathrm{M}}(m)^n.$$

There are only finitely many pairs $1 \leqslant m \leqslant j$ and $0 \leqslant d \leqslant j$, so these bounds hold with a common constant $C = C(\mathrm{M}, j)$. Therefore,

$$\begin{aligned}\left|\sum_{m=1}^{j} \big(Q_{m,j+1-m}(\mathrm{M}^{\oplus n}) - Q_{m,j-m}(\mathrm{M}^{\oplus n})\big)\right| &\leqslant C \sum_{m=1}^{j} \big(n^{j+1-m} + n^{j-m}\big) Z_{\mathrm{M}}(m)^n \\ &\leqslant 2jC\, n^j Z_{\mathrm{M}}(j)^n.\end{aligned}$$

Updating the bounding constant C gives the desired statement. □

Remark 2.8 The work of [Pie25] gives a decomposition of Chow rings of direct sums of possibly different matroids using the Chow rings of the matroid summands. It would be interesting to obtain our asymptotic formula above using the decomposition.

Proof of Theorem A. Suppose that for any $j \geqslant 2$, $Z_{\mathrm{M}}(j)^2 < Z_{\mathrm{M}}(j-1)Z_{\mathrm{M}}(j+1)$. By Proposition 2.6, for a fixed $j \geqslant 1$, as n goes to infinity, the jth coefficient $P_j(\mathrm{M}^{\oplus n})$ is asymptotically equal to $Z_{\mathrm{M}}(j+1)^n$. Furthermore,

$$P_0(\mathrm{M}^{\oplus n}) = 1 = Z_{\mathrm{M}}(1)^n.$$

Therefore, for any $j \geqslant 2$, the asymptotic ratio

$$\lim_{n\to\infty} \frac{P_{j-1}(\mathrm{M}^{\oplus n})^2}{P_{j-2}(\mathrm{M}^{\oplus n})P_j(\mathrm{M}^{\oplus n})} = \lim_{n\to\infty} \left(\frac{Z_{\mathrm{M}}(j)^2}{Z_{\mathrm{M}}(j-1)Z_{\mathrm{M}}(j+1)}\right)^n = 0 < 1,$$

which implies that log-concavity of coefficients of $P(\mathrm{M}^{\oplus n})$ fails for sufficiently large n. Newton's inequalities in turn imply failures of real-rootedness [New07, Brä15]. □

2.5. **Log-concavity and real-rootedness.** We briefly recall relevant properties of polynomials, and refer the reader to Brändén's survey [Brä15]. A polynomial $p(t) = \sum_{i=0}^{d} a_i t^i \in \mathbb{R}_{\geqslant 0}[t]$ is *log-concave*, if $a_i^2 \geqslant a_{i-1}a_{i+1}$ for every i. For every i, the *defect at index* i is defined to be $a_i^2 - a_{i-1}a_{i+1}$. If $p(t)$ only has real zeros, the well-known result of Newton [New07, pp. 242–245] implies *ultra log-concavity* of the coefficients of $p(t)$:

$$\frac{a_i^2}{\binom{d}{i}^2} \geqslant \frac{a_{i-1}}{\binom{d}{i-1}} \frac{a_{i+1}}{\binom{d}{i+1}}, \qquad \text{for every } 0 < i < d,$$

which implies log-concavity with no internal zeros.

If a polynomial $p(t) = \sum_{i=0}^{d} a_i t^i \in \mathbb{R}_{\geqslant 0}[t]$ is palindromic, it admits a unique expansion into the basis $\{t^i(1+t)^{d-2i}\}_{i \geqslant 0}$, and has γ-polynomial

$$p(t) = \sum_i \gamma_i \, t^i (1+t)^{d-2i}, \quad \gamma(p(t); x) \coloneqq \sum_{i=0}^{\lfloor d/2 \rfloor} \gamma_i x^i.$$

Since the Chow ring of a matroid is a Poincaré duality algebra, the Chow polynomial is palindromic. By [Stu24, FMV24], its γ-coefficients can be written using the flag h-vector as follows:

$$\underline{\mathrm{H}}(\mathrm{M}; t) = \sum_i \gamma_i t^i (1+t)^{r-1-2i}, \quad \gamma_i = \sum_{\substack{S \subseteq [r-1] \text{ stable} \\ 1 \notin S,\ |S| = i}} \beta_{\mathrm{M}}(S). \tag{10}$$

We will be particularly interested in Chow polynomials of degree 6. For any matroid of rank 7, the only stable subset of $\{2, \ldots, 6\}$ with size 3 is $\{2, 4, 6\}$. Therefore,

$$\gamma_0 = 1, \qquad \gamma_3 = \beta_{\mathrm{M}}(246),$$

and the coefficients of the Chow polynomial satisfy

$$\begin{aligned} \underline{\mathrm{H}}_0 &= \underline{\mathrm{H}}_6 = 1, \\ \underline{\mathrm{H}}_1 &= \underline{\mathrm{H}}_5 = \gamma_1 + \binom{6}{1}, \\ \underline{\mathrm{H}}_2 &= \underline{\mathrm{H}}_4 = \gamma_2 + \binom{4}{1}\gamma_1 + \binom{6}{2}, \\ \underline{\mathrm{H}}_3 &= \gamma_3 + \binom{2}{1}\gamma_2 + \binom{4}{2}\gamma_1 + \binom{6}{3}. \end{aligned} \tag{11}$$

Lemma 2.9 *Let* M *be a loopless matroid of rank* 7*.*

(a) $\underline{\mathrm{H}}(\mathrm{M}; t)$ *is log-concave if and only if*

$$\Delta_{\underline{\mathrm{H}}} \coloneqq \underline{\mathrm{H}}_2^2 - \underline{\mathrm{H}}_1 \underline{\mathrm{H}}_3 \geqslant 0. \tag{12}$$

(b) $\underline{\mathrm{H}}(\mathrm{M}; t)$ *is real-rooted if and only if the discriminant for the cubic polynomial* $\gamma(\underline{\mathrm{H}}(t); x)$ *is nonnegative:*

$$\operatorname{disc}(\gamma(\underline{\mathrm{H}}(t); x)) = \gamma_1^2\gamma_2^2 - 4\gamma_1^3\gamma_3 - 4\gamma_2^3 + 18\gamma_1\gamma_2\gamma_3 - 27\gamma_3^2 \geqslant 0.$$

Proof. The first statement follows from the definition and palindromicity of $\underline{\mathrm{H}}(\mathrm{M}; t)$. The second statement follows from the equivalence of real-rootedness of a polynomial and its γ-polynomial. See, for instance, [Pet15, Observation 4.2]. □

3. Matroid operations: hyperplane coalescence, absorption

We study the properties of two crucial matroid operations, which are essential in our construction: hyperplane coalescence, and a special class of extensions by modular cuts, called *absorption*. For background on matroids, we refer to [Oxl11].

We start with the classical theory of detruncation, introduced by Crapo [Cra70] to construct matroids with prescribed hyperplanes and a fixed truncation. See also [PvdP17, Section 3 and Theorem 12].

Definition 3.1 Let M be a matroid of positive rank, and $k \geqslant 0$. The kth truncation $\mathrm{T}^k(\mathrm{M})$ of M is the matroid with rank $r-k$ with lattice of flats

$$\mathscr{L}(\mathrm{T}^k(\mathrm{M})) = \{F \in \mathscr{L}(\mathrm{M})\colon \mathrm{rk}_{\mathrm{M}}(F) < k\} \cup \{E\}.$$

If $k = 1$, the truncation $\mathrm{T}^k(\mathrm{M})$ is called the *principal truncation*, or simply *truncation*. A matroid M of rank r is a *detruncation* of a matroid T of rank $r-1$, if the principal truncation $\mathrm{Tr}(\mathrm{M})$ is T.

The following properties are standard.

Lemma 3.2 *For a matroid* M *of rank* r *on* E *and* $1 \leqslant k \leqslant r$*, the following hold.*

(a) The flats of $\mathrm{Tr}^k(\mathrm{M})$ *are the flats of* M *of rank at most* $k-1$*, with their rank in* M*, and* E*. In particular* $\alpha_{\mathscr{L}(\mathrm{Tr}^k(\mathrm{M}))}(S) = \alpha_{\mathscr{L}(\mathrm{M})}(S)$ *for every* $S \subseteq [k-1]$ *(*[Oxl11, Proposition 7.3.10]*).*

(b) If M *is loopless and* $k < r$*, then* $\mathrm{Tr}^k(\mathrm{M})$ *is connected (*[Oxl11, Section 7.3, Exercise 8]*).*

(c) If M *is simple and* $k \geqslant 2$*, then* $\mathrm{Tr}^k(\mathrm{M})$ *is simple (*[Oxl11, Proposition 7.3.10]*).*

3.1. **Hyperplane coalescence.** Hyperplane coalescence is a natural geometric operation on matroids that merges groups of hyperplanes into larger hyperplanes while preserving the geometry of lower ranks. It is an application of Crapo's criterion in the Crapo–Knuth theory of detruncations [Cra70, Knu75, PvdP17]. The spanning subsets that become new hyperplanes are called *blocks*; the lemma below specifies which blocks may be selected. For a single block with restriction $\mathrm{U}_{r,|H|}$, coalescence replaces this restriction by $\mathrm{U}_{r-1,|H|}$, reversing stressed-hyperplane relaxation [FNV22, Theorem 3.4 and Proposition 3.10]. When $|H| = r$, this is tightening a free basis.

Definition 3.3 (Hyperplane coalescence) Let M be a matroid of rank $r \geqslant 2$ on E, and let $\mathscr{H}$ be a family of proper spanning subsets of E. A matroid M' of rank r on E is obtained from M via *hyperplane coalescence along* $\mathscr{H}$ if:

(a) M and M' have the same flats of rank at most $r-2$, with the same ranks;

(b) the hyperplanes of M' are precisely the elements in $\mathscr{H}$ together with the hyperplanes of M not contained in any element in $\mathscr{H}$.

In other words, the collection of hyperplanes of M' is

$$\mathscr{L}^{r-1}(\mathrm{M}') = \mathscr{H} \ \cup \ \big\{F \in \mathscr{L}^{r-1}(\mathrm{M}) : F \not\subseteq H \text{ for every } H \in \mathscr{H}\big\}.$$

The members of $\mathscr{H}$ are called *blocks*. We say that M' is obtained by *coalescing the blocks* $\mathscr{H}$ *in* M. Thus, to *coalesce* a block H is to replace the hyperplanes contained in H by the single hyperplane H. For each block H, the hyperplanes in M contained in H are replaced by the single hyperplane H in M'.

Example 3.4 For $\mathrm{U}_{3,4}$, the coalescence along $\mathscr{H} = \{123\}$ of $\mathrm{U}_{3,4}$ is $\mathrm{U}_{2,3} \oplus \mathrm{U}_{1,1}$. See Figure 1.

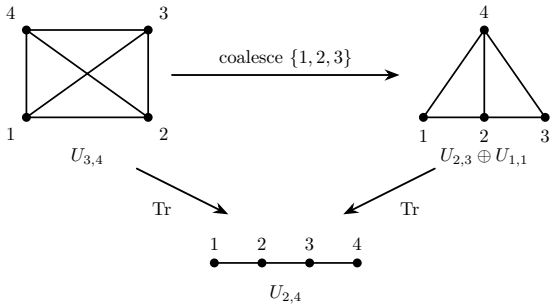


FIGURE 1.

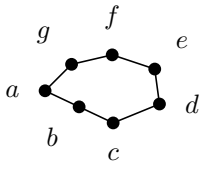
The following example shows that hyperplane coalescence need not always exist for an arbitrary collection $\mathscr{H}$ of proper spanning subsets of E.

Example 3.5 Let $\mathrm{M} = \mathrm{U}_{3,4}$ on $E = [4]$, and consider

$$\mathscr{H} = \{123, 124\}.$$

Both members of $\mathscr{H}$ are proper subsets of full rank 3. However, hyperplane coalescence along $\mathscr{H}$ does not exist. Indeed, any such matroid M' would be simple, since its flats of rank 1 are the same as those of M. The two blocks would then be distinct flats of rank 2 containing the independent pair $\{1, 2\}$. Both would therefore equal $\mathrm{cl}_{\mathrm{M}'}(\{1, 2\})$, a contradiction.

Example 3.6 In Crapo's original work on detruncations, there is an example of all 13 detruncations of the following combinatorial geometry of rank 3.



The *free detruncation* N of a matroid M of rank r is a matroid of rank $r + 1$ such that for every detruncation N' of M, the independent sets of N contain those of N':

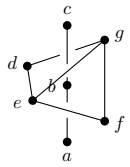
$$\mathcal{I}(N') \subseteq \mathcal{I}(N).$$

Figure 2 shows all the detruncations of Crapo's geometry of rank 3, with its free detruncation on top. All the detruncations can be viewed as hyperplane coalescences of the free detruncation along a set of blocks $\mathscr{H}$.

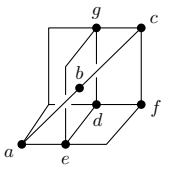
The following lemma gives a sufficient condition on $\mathscr{H}$ for a coalescence to exist, and when such a matroid exists, it is unique.

Lemma 3.7 *Let* M *have rank* $r \geqslant 2$ *on* E*, and let* $\mathscr{H}$ *be a family of proper spanning subsets satisfying*

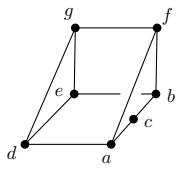
Crapo's rank-four detruncations



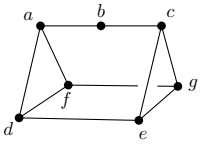
$F = \mathrm{Tr}^4(U_{2,3} \oplus U_{4,4}), \qquad \mathscr{H} = \varnothing$



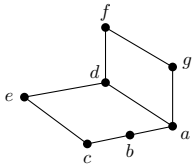
$\mathscr{H} = \{adef, bdeg, cdfg\}$



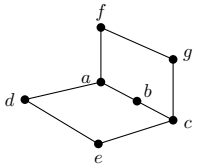
$\mathscr{H} = \{abcde, adfg, befg\}$



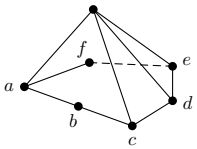
$\mathscr{H} = \{abcde, abcfg, defg\}$



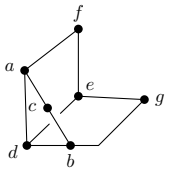
$\mathscr{H} = \{abcde, adfg\}$



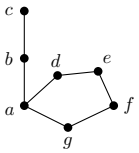
$\mathscr{H} = \{abcde, abcfg\}$



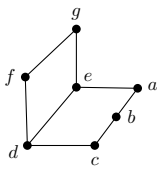
$\mathscr{H} = \{abcdef\}$



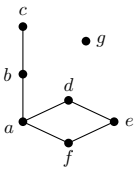
$\mathscr{H} = \{adef, bdeg\}$



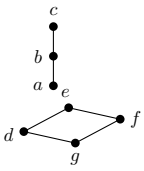
$\mathscr{H} = \{adefg\}$



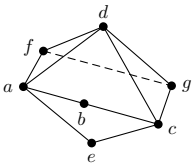
$\mathscr{H} = \{abcde, defg\}$



$\mathscr{H} = \{adef\}$



$\mathscr{H} = \{defg\}$



$\mathscr{H} = \{abcde\}$

FIGURE 2.

(a) $\mathrm{rk}_{\mathrm{M}}(H \cap H') \leqslant r - 2$ *for distinct* $H, H' \in \mathscr{H}$*;*
(b) $\mathrm{cl}_{\mathrm{M}}(I) \subseteq H$ *whenever* $I \subseteq H \in \mathscr{H}$ *is independent and* $|I| = r - 1$.

Then $\mathscr{H}$ *uniquely determines a hyperplane coalescence* M' *of* M*, which has bases*

$$\mathscr{B}(\mathrm{M}') = \{B \in \mathscr{B}(\mathrm{M}) \colon B \nsubseteq H \in \mathscr{H}\}.$$

and rank function

$$\mathrm{rk}_{\mathrm{M}'}(S) = \begin{cases} r - 1, & \mathrm{rk}_{\mathrm{M}}(S) = r \text{ and } S \subseteq H \text{ for some } H \in \mathscr{H}, \\ \mathrm{rk}_{\mathrm{M}}(S), & \text{otherwise.} \end{cases} \tag{13}$$

Proof. This follows from Crapo's criterion [PvdP17, Theorem 12] applied to Tr(M). □

One advantage of the hyperplane coalescence is that it changes multichain counts in a controlled way, enabling us to apply Theorem A. Since only the hyperplanes change under coalescence, the change in the zeta polynomial is a sum of contributions from the individual blocks. Let M be loopless of rank $r \geqslant 2$, let $\mathscr{H}$ be a collection of blocks satisfying Lemma 3.7, and let M′ be its coalescence along $\mathscr{H}$.

Proposition 3.8 *For every $m \geqslant 0$,*

$$Z_{\mathrm{M}'}(m) = Z_{\mathrm{M}}(m) - \sum_{H \in \mathscr{H}} \left(Z_{\mathrm{M}^H}(m) - \sum_{i=0}^{m} Z_{\mathrm{Tr}(\mathrm{M}^H)}(i) \right). \tag{14}$$

This will let us compute the multichain counts of our geometric families without enumerating their lattices. To prove the proposition, we count the affected flags.

Lemma 3.9 *For every $S \subseteq [r-2]$,*

$$\begin{aligned} \alpha_{\mathrm{M}'}(S) &= \alpha_{\mathrm{M}}(S), \\ \alpha_{\mathrm{M}'}(S \cup \{r-1\}) &= \alpha_{\mathrm{M}}(S \cup \{r-1\}) + \sum_{H \in \mathscr{H}} \big(\alpha_{\mathrm{M}^H}(S) - \alpha_{\mathrm{M}^H}(S \cup \{r-1\})\big). \end{aligned} \tag{15}$$

Proof. This follows from the fact that flats of rank less than $r-1$ are the same for M and M′, together with the closure hypothesis in Lemma 3.7. □

Lemma 3.10 *For $P = f$ or $P = \underline{\mathrm{H}}$,*

$$P(\mathrm{M}'; t) = P(\mathrm{M}; t) - \sum_{H \in \mathscr{H}} \big(P(\mathrm{M}^H; t) - (1+t)P(\mathrm{Tr}(\mathrm{M}^H); t)\big). \tag{16}$$

Proof. Summing (15) by chain length gives the assertion for f.

For the Chow polynomial, the weights in (6) satisfy, for $S \subseteq [r-2]$,

$$\underline{\tau}(S \cup \{r\}) = t\big(\underline{\tau}(S) + \underline{\tau}(S \cup \{r-1\})\big), \qquad \underline{\tau}(S \cup \{r-1, r\}) = 0.$$

Thus (6) implies that, for any loopless matroid M of rank r,

$$\underline{\mathrm{H}}(\mathrm{M}; t) - (1+t)\underline{\mathrm{H}}(\mathrm{Tr}(\mathrm{M}); t) = \sum_{S \subseteq [r-2]} \big(\alpha_{\mathrm{M}}(S \cup \{r-1\}) - \alpha_{\mathrm{M}}(S)\big)\underline{\tau}(S \cup \{r-1\}).$$

Applying (15) with these weights proves the result. □

Proof of Proposition 3.8. The binomial transform (3) sends t^j to $\binom{m}{j+1}$. Its value on $(1+t)t^j$ is

$$\binom{m}{j+1} + \binom{m}{j+2} = \binom{m+1}{j+2} = \sum_{a=0}^{m} \binom{a}{j+1}.$$

The assertion follows by applying this transform to (16) with $P = f$. □

3.2. **Absorption.** In this subsection, we use hyperplane coalescence to construct extensions of a matroid by adjoining auxiliary points, each lying in the closure of every hyperplane in a prescribed family. We call this construction *absorption*. We first adjoin the auxiliary points as coloops and truncate back to the original rank. The union of each selected hyperplane with its assigned point is then a spanning subset. Under the conditions below, these subsets can be coalesced simultaneously, placing each auxiliary point in the required closures. We use this construction for the affine and Gabidulin families.

Proposition 3.11 *Let* G *have rank* $r \geqslant 2$*, and let* $\mathscr{F}_1, \dots, \mathscr{F}_m$ *be pairwise disjoint families of hyperplanes such that*

$$\mathrm{rk}_{\mathrm{G}}(F \cap F') \leqslant r-3 \qquad (F, F' \in \mathscr{F}_i,\ F \neq F').$$

For a set of new points $X = \{p_1, \dots, p_m\}$*, put*

$$\mathrm{M} = \mathrm{Tr}^r(\mathrm{G} \oplus \mathrm{U}_{X,X}), \qquad \mathscr{H} = \{F \cup \{p_i\} : F \in \mathscr{F}_i,\ 1 \leqslant i \leqslant m\}.$$

Then the blocks in $\mathscr{H}$ *can be coalesced in* M*. The resulting matroid* M' *restricts to* G *and, for* $A \subseteq E(\mathrm{G})$ *and* $J \subseteq X$*, has rank*

$$\mathrm{rk}_{\mathrm{M}'}(A \cup J) = \begin{cases} r-1, & J = \{p_i\} \text{ and } \mathrm{cl}_{\mathrm{G}}(A) \in \mathscr{F}_i, \\ \min\{\mathrm{rk}_{\mathrm{G}}(A) + |J|, r\}, & \text{otherwise.} \end{cases} \tag{17}$$

Before coalescence, each block has restriction

$$\mathrm{M}|(F \cup \{p_i\}) = (\mathrm{G}|F) \oplus \mathrm{U}_{1,1}.$$

For each $F \in \mathscr{F}_i$, the old hyperplanes F and $A \cup \{p_i\}$, with A a hyperplane of $\mathrm{G}|F$, all acquire closure $F \cup \{p_i\}$. The rank formula also shows that the order of adjoining the auxiliary points does not affect the result.

Proof. Put $\mathrm{Q} = \mathrm{G} \oplus \mathrm{U}_{X,X}$. Each block $H = F \cup \{p_i\}$ is a proper rank-r flat of Q, hence a proper spanning subset of M. For every independent $(r-1)$-set $I \subseteq H$,

$$\mathrm{cl}_{\mathrm{M}}(I) = \mathrm{cl}_{\mathrm{Q}}(I) \subseteq H.$$

Distinct blocks sharing p_i intersect in rank at most $(r-3)+1 = r-2$. Blocks with different auxiliary points intersect in rank at most $r-2$, since their geometric hyperplanes are distinct. Thus both conditions of Lemma 3.7 hold.

A set $A \cup J$ is spanning in M and contained in a block $F \cup \{p_i\}$ exactly when $J = \{p_i\}$ and $\mathrm{cl}_{\mathrm{G}}(A) = F$. Hence (13) gives (17); taking $J = \varnothing$ shows that M' restricts to G. The old block restriction follows because truncation to rank r does not change the restriction to a rank-r subset. □

4. Four families from finite geometries

We construct the affine Klein family, the affine packing family, the Gabidulin family, and the projective family from finite geometries, and prove Theorem B.

4.1. **The affine family.** Let q be a power of an odd prime. Recall that the affine geometry $\mathrm{AG}(4,q)$ is the matroid of rank 5, which has flats given by affine subspaces of $\mathbb{F}_q^4$, and the projective geometry $\mathrm{PG}(3,q)$ is the matroid of rank 4, which has flats given by projective subspaces of $\mathbb{P}(\mathbb{F}_q^4)$. We will construct the affine family of matroids M_q using classical operations from projective incidence geometry and combinatorial design, in four steps:

(i) Start with the truncation $\mathrm{Tr}(\mathrm{AG}(4,q))$.

(ii) Use the Klein correspondence to choose pairwise disjoint spreads of $\mathrm{PG}(3,q)$. Each spread specifies a family of affine planes through their directions.

(iii) Construct the matroid inductively: for each chosen spread, adjoin a new coloop to the current matroid, truncate once, and absorb the new point into the selected planes. Define M_q as the resulting matroid.

(iv) Compute flag vectors of M_q, and prove that they only depend on q.

For any field $\mathbb{F}$ and a vector space V over $\mathbb{F}$, the *direction* of a nonempty affine subspace $A \subseteq V$ is the linear subspace of V

$$D(A) \coloneqq \{x - y : x, y \in A\} \subseteq V.$$

For example, the direction $D(L)$ is equal to L for any linear subspace $L \subseteq V$; and the direction of any affine translation $v + L$ of a linear space L for $v \in V$ is L. The projectivization of the direction of an affine subspace of dimension k is a projective subspace of dimension $k-1$. Therefore, for each flat F of rank k in $\mathrm{AG}(n, q)$, $\mathbb{P}(D(F))$ is a flat of rank $k-1$ in $\mathrm{PG}(n-1, q)$.

A *k-spread* of $\mathrm{PG}(n, q)$ is a collection $\mathcal{S}$ of projective k-dimensional subspaces such that every projective point belongs to exactly one element of $\mathcal{S}$. A 1-spread is called a *line spread*, or simply *spread*. Two k-spreads are *disjoint* if they contain no common points. A k-spread of $\mathrm{PG}(n, q)$ exists if and only if $k+1$ divides $n+1$, and every such spread has cardinality

$$|\mathcal{S}| = \frac{q^{n+1} - 1}{q^{k+1} - 1}.$$

For $\mathrm{PG}(3, q)$, every spread $\mathcal{S}$ consists of q^2+1 projective lines. For general background on spreads in finite projective spaces, see [Dem68, Hir98, Etz15].

Given a spread $\mathcal{S}$, we consider the collection of all affine planes whose projectivized directions are elements in $\mathcal{S}$

$$\mathscr{A}(\mathcal{S}) \coloneqq \{A \subseteq V : \mathbb{P}(D(A)) \in \mathcal{S}\}.$$

Since each direction in $\mathcal{S}$ admits q^2 affine translates,

$$|\mathscr{A}(\mathcal{S})| = q^2(q^2 + 1).$$

The following lemma describes the intersections among the elements in $\mathscr{A}(\mathcal{S})$.

Lemma 4.1 *Let V be a vector space over a field, and let $P = a + D$ and $P' = b + D'$ be distinct affine subspaces of V.*

(a) If $D = D'$, then $P \cap P' = \varnothing$.

(b) If $D \cap D' = \{0\}$, then $P \cap P'$ contains at most one point.

(c) If $V = D \oplus D'$, then $P \cap P'$ contains exactly one point.

In particular, if $V = \mathbb{F}_q^{2k+1}$ and $\mathbb{P}(D), \mathbb{P}(D')$ belong to a k-spread of $\mathrm{PG}(2k, q)$, then the distinct affine k-dimensional subspaces P and P' are either parallel and disjoint, or have complementary directions and intersect in exactly one point.

Proof. Since a, b are distinct vectors, (a) follows immediately. For (b), if $x, y \in P \cap P'$, then $x - y \in D \cap D' = \{0\}$, and $x = y$. For (c), write $b - a = d + d'$ with $d \in D$ and $d' \in D'$. Then $a + d = b - d'$ belongs to $P \cap P'$, and (b) implies its uniqueness. □

Example 4.2 Let $V = \mathbb{F}_q^4$ and let $\mathcal{S}$ be a spread of $\mathrm{PG}(3, q)$. Let two distinct affine subspaces $P = a + D, P' = b + D'$ of V be given by distinct directions $\mathbb{P}(D) \neq \mathbb{P}(D')$ in $\mathcal{S}$. Since elements in $\mathcal{S}$ do not share common points, Lemma 4.1 implies that two distinct affine planes P, P' in $\mathscr{A}(\mathcal{S})$ are either parallel and disjoint, or meet in exactly one point. See Figure 3.

Next, we use classical Plücker embeddings of Grassmannians to construct $\frac{q^2+1}{2}$ pairwise disjoint spreads of $\mathrm{PG}(3, q)$. The Plücker embedding realizes the Grassmannian $\mathrm{Gr}(2, V)$ of two-dimensional linear subspaces of an n-dimensional vector space

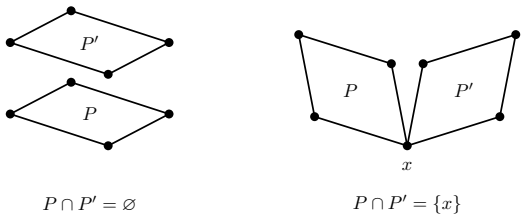


FIGURE 3.

V as a subvariety of $\mathbb{P}^{\binom{n}{2}-1} \cong \mathbb{P}(\bigwedge^2 V)$:

$$\iota\colon \mathrm{Gr}(2,V) \hookrightarrow \mathbb{P}\Big(\bigwedge^2 V\Big), \qquad \mathrm{span}(u,v) \longmapsto [u \wedge v].$$

When $n = 4$, the image is defined by a single quadratic equation, given by the Plücker relations. This special case is classically known as the *Klein correspondence*, and its image as the *Klein quadric*, studied by Klein in his 1868 dissertation, republished in [Kle84]. Over finite fields, the Klein correspondence identifies regular spreads with elliptic sections of the Klein quadric. For usage of Klein correspondence to construct regular spreads in the existing literature, see [Hir85, Section 15.4 and Chapter 17]. Let $V = \mathbb{F}_q^4$ with basis $v_1, \ldots, v_4$, and denote

$$v_{ij} \coloneqq v_i \wedge v_j \in \bigwedge^2 V, \qquad W \coloneqq \bigwedge^2 V$$

The Plücker relation defines the quadratic form on W with its bilinear form

$$Q\colon W \to \mathbb{F}_q, \quad x \mapsto x_{12}x_{34} - x_{13}x_{24} + x_{14}x_{23}$$
$$B\colon W^2 \to \mathbb{F}_q, \quad (x,y) \mapsto Q(x+y) - Q(x) - Q(y).$$

The *Klein quadric* is the image of the Plücker embedding, or equivalently, the vanishing locus of the quadratic form Q

$$\mathscr{K}\colon \{\, [x] \in \mathbb{P}(W) : Q(x) = 0 \,\},$$

and the Plücker embedding gives the set bijection

$$\kappa : \mathscr{L}_2(\mathrm{PG}(3,q)) \longrightarrow \mathscr{K}, \qquad \mathbb{P}(\mathrm{span}(u,v)) \longmapsto [u \wedge v].$$

In addition, we also have the set bijection by definition,

$$\gamma\colon \mathscr{L}_2(\mathrm{PG}(3,q)) \xrightarrow{\simeq} \mathrm{Gr}(2,V).$$

Therefore, there is a commutative diagram of set morphisms

$$\begin{array}{ccc} \mathscr{L}_2(\mathrm{PG}(3,q)) & \xrightarrow{\qquad\gamma\qquad} & \mathrm{Gr}(2,V) \\ & {\scriptstyle\kappa}\searrow \quad \swarrow{\scriptstyle\iota} & \\ & \mathbb{P}\big(\bigwedge^2 V\big) \cong \mathbb{P}^5 & \end{array} \tag{18}$$

A quadric in $\mathbb{P}^3$ is *elliptic* if it is smooth, and contains no projective lines. An *elliptic section* of the Klein quadric $\mathscr{K}$ is an intersection with a $\mathbb{P}^3$ which contains no projective lines. Under the Klein correspondence, elliptic sections correspond precisely to regular spreads of $\mathrm{PG}(3,q)$. Therefore, our task reduces to constructing $\frac{q^2+1}{2}$ pairwise disjoint elliptic sections of $\mathscr{K}$.

Lemma 4.3 *The Klein correspondence*

$$\kappa : \mathscr{L}_2(\mathrm{PG}(3,q)) \longrightarrow \mathscr{K}$$

induces a bijection between elliptic sections of $\mathscr{K}$ *and regular spreads of* $\mathrm{PG}(3,q)$*, given by*

$$\mathscr{C} \longmapsto \mathcal{S}(\mathscr{C}) := \kappa^{-1}(\mathscr{C}) \subseteq \mathscr{L}_2(\mathrm{PG}(3,q)).$$

Proof. Let $\mathscr{C} = \mathscr{K} \cap \mathbb{P}(U)$ be an elliptic section. If two distinct lines in $\mathcal{S}(\mathscr{C})$ intersected, the projective line joining their Klein images would lie entirely on $\mathscr{K}$. It would also lie in $\mathbb{P}(U)$, hence in $\mathscr{C}$, contradicting ellipticity. Thus the projective lines of $\mathcal{S}(\mathscr{C})$ are pairwise skew. An elliptic quadric has q^2+1 points, so $\mathcal{S}(\mathscr{C})$ consists of q^2+1 pairwise skew lines. Since each projective line contains $q+1$ points, and

$$(q^2+1)(q+1) = q^3+q^2+q+1 = |\mathrm{PG}(3,q)|,$$

these lines partition the point set of $\mathrm{PG}(3,q)$ and form a spread. The fact that this spread is regular, and that every regular spread arises from an elliptic section, is the classical regular-spread correspondence [Hir85, Section 15.4 and Chapter 17]. Furthermore, since κ is a bijection, the regular spread from an elliptic section is unique. □

To construct $\frac{q^2+1}{2}$ pairwise disjoint elliptic sections of $\mathscr{K}$, we use *polarity* given by the nondegenerate bilinear form B on $\mathscr{K}$. A projective line $\ell \subseteq \mathbb{P}(W)$ is *external* to $\mathscr{K}$, if $\ell \cap \mathscr{K} = \varnothing$. The quadratic form Q has associated nondegenerate symmetric bilinear form

$$B(x,y) = Q(x+y) - Q(x) - Q(y).$$

For a linear subspace $A \subseteq W$, we define its orthogonal complement by

$$A^{\perp_B} \coloneqq \{w \in W : B(w,a) = 0 \text{ for every } a \in A\}.$$

The *polar* of the projective subspace $\mathbb{P}(A)$ is the projectivization of this complement

$$\mathbb{P}(A)^{\perp} \coloneqq \mathbb{P}(A^{\perp_B}).$$

Since W has dimension $\binom{4}{2} = 6$, the polar of a projective line is a $\mathbb{P}^3$. Moreover, if $\langle S,T\rangle$ denotes the projective span of two projective subspaces, then

$$(S^{\perp})^{\perp} = S, \qquad \langle S,T\rangle^{\perp} = S^{\perp} \cap T^{\perp}.$$

For the Klein quadric $\mathscr{K}$, the polar of an external line meets $\mathscr{K}$ in an elliptic section. Thus every external line ℓ determines an elliptic quadric $\mathscr{K} \cap \ell^{\perp}$ and hence a regular spread of $\mathrm{PG}(3,q)$. We will construct several external lines so that their polar projective spaces meet only outside $\mathscr{K}$.

Construction 4.4 Let q be an odd prime power. Choose an external line

$$\ell_0 = \mathbb{P}(A_0) \subseteq \mathbb{P}(W) \qquad \text{with } \dim A_0 = 2,$$

and define the orthogonal complement and its associated elliptic section of $\mathscr{K}$,

$$U = A_0^{\perp_B}, \qquad \mathscr{C}_0 = \mathscr{K} \cap \mathbb{P}(U).$$

Then $\dim U = 4$, $\mathbb{P}(U) = \ell_0^{\perp}$, and $\mathscr{C}_0$ is an elliptic quadric. By Lemma 4.6, there exists a fixed-point-free projective involution τ of $\mathbb{P}(U)$ satisfying

$$\tau^2 = \mathrm{id}, \qquad \tau(x) \neq x \quad \text{for every } x \in \mathbb{P}(U), \qquad \tau(\mathscr{C}_0) = \mathscr{C}_0.$$

Let $\{\ell_1, \dots, \ell_m\}$ denote the distinct lines spanned by $\{x, \tau(x)\}$ for $x \in \mathbb{P}(U)$ which are disjoint from $\mathscr{C}_0$. These lines are external to $\mathscr{K}$, since they lie in $\mathbb{P}(U)$ and are disjoint from $\mathscr{C}_0 = \mathscr{K} \cap \mathbb{P}(U)$. We therefore define for each $1 \leqslant i \leqslant m$,

$$\mathscr{C}_i \coloneqq \mathscr{K} \cap \ell_i^{\perp}, \qquad \mathcal{S}_i \coloneqq \kappa^{-1}(\mathscr{C}_i).$$

Lemma 4.5 *Construction 4.4 gives $\frac{q^2+1}{2}$ pairwise disjoint elliptic sections of $\mathscr{K}$.*

Proof. Each line $\operatorname{span}(x, \tau(x))$ is preserved by τ. Two distinct such lines cannot meet, since their unique intersection point would be fixed by τ. Every point lies on its joining line, so these lines form a spread of $\mathbb{P}(U)$, consisting of q^2+1 lines. Each of these lines meets $\mathscr{C}_0$ in either zero or two points. Indeed, a line containing $x \in \mathscr{C}_0$ also contains the distinct point $\tau(x) \in \mathscr{C}_0$; it cannot contain more than two points of the quadric without lying entirely on it. Since $\mathscr{C}_0$ has q^2+1 points, exactly half the spread lines meet it, and the other half avoid it. Therefore $m = \frac{q^2+1}{2}$.

For $i \neq j$, the lines ℓ_i, ℓ_j are skew and hence span $\mathbb{P}(U)$. Therefore,

$$\ell_i^{\perp} \cap \ell_j^{\perp} = \langle \ell_i, \ell_j \rangle^{\perp} = \mathbb{P}(U)^{\perp} = \ell_0.$$

Their polar spaces meet precisely along the external line ℓ_0. Hence

$$\mathscr{C}_i \cap \mathscr{C}_j = \mathscr{K} \cap \ell_0 = \varnothing.$$

The corresponding regular spreads $\mathcal{S}_i, \mathcal{S}_j$ therefore share no common elements. □

Lemma 4.6 *For an odd prime power q, the projective 3-space $\mathbb{P}(U)$ in Construction 4.4 admits a fixed-point-free projective involution preserving $\mathscr{C}_0$.*

Proof. In diagonal coordinates, we write $Q = Q_1 \oplus Q_2$, where

$$Q_i(s,t) = a_i s^2 + b_i t^2, \qquad a_i, b_i \neq 0.$$

Fix a nonsquare $d \in \mathbb{F}_q$. For each i, the sets of values u^2 and $d - (a_i/b_i)v^2$ each have $(q+1)/2$ elements, so they intersect. Thus we can choose u_i, v_i satisfying

$$u_i^2 + \frac{a_i}{b_i} v_i^2 = d.$$

The matrices

$$J_i = \begin{pmatrix} u_i & v_i \\ (a_i/b_i)v_i & -u_i \end{pmatrix}$$

satisfy $J_i^2 = dI$ and $Q_i(J_i x) = dQ_i(x)$. Therefore, $J = J_1 \oplus J_2$ satisfies

$$J^2 = dI, \qquad Q(Jx) = dQ(x).$$

Its projectivization $\tau([x]) = [Jx]$ is an involution preserving $\mathscr{C}_0$. A fixed point would give $Jx = \lambda x$ for some $x \neq 0$ and $\lambda \in \mathbb{F}_q$ with $\lambda^2 = d$, a contradiction. □

We now associate a matroid with any collection of pairwise disjoint spreads. At each step, we adjoin one coloop, truncate once, and perform the prescribed absorption.

Definition 4.7 Let q be a prime power, let $V = \mathbb{F}_q^4$, and let $\mathcal{S}_1, \dots, \mathcal{S}_m$ be pairwise disjoint spreads of $\mathrm{PG}(3,q)$. Choose distinct new elements $p_1, \dots, p_m$ outside V. Define matroids $\mathrm{M}_0, \dots, \mathrm{M}_m$ inductively as follows. Set

$$\mathrm{M}_0 \coloneqq \mathrm{Tr}^1(\mathrm{AG}(4,q)).$$

For $1 \leqslant i \leqslant m$, let

$$\widetilde{\mathrm{M}}_i \coloneqq \mathrm{Tr}^1\big(\mathrm{M}_{i-1} \oplus \mathrm{U}_{\{p_i\},\{p_i\}}\big), \qquad \mathscr{H}_i \coloneqq \{P \cup \{p_i\} : P \in \mathscr{A}(\mathcal{S}_i)\}.$$

Let M_i be the hyperplane coalescence of $\widetilde{\mathrm{M}}_i$ along $\mathscr{H}_i$, and let

$$\mathrm{M}(\mathcal{S}_1, \ldots, \mathcal{S}_m) = \mathrm{M}_m.$$

Definition 4.8 (The affine Klein family) For odd q, let $\{\mathcal{S}_1, \ldots, \mathcal{S}_{(q^2+1)/2}\}$ be the collection of spreads given by Construction 4.4. The *affine Klein family* is defined by

$$\mathrm{M}_q \coloneqq \mathrm{M}(\mathcal{S}_1, \ldots, \mathcal{S}_{(q^2+1)/2}).$$

Proposition 4.9 *Let q be a prime power, let $V = \mathbb{F}_q^4$, and let $\mathcal{S}_1, \ldots, \mathcal{S}_m$ be pairwise disjoint spreads of $\mathrm{PG}(3,q)$. Let*

$$X = \{p_1, \ldots, p_m\}, \qquad E = V \sqcup X, \qquad \mathscr{A}_i = \mathscr{A}(\mathcal{S}_i).$$

Then $\mathrm{M}(\mathcal{S}_1, \ldots, \mathcal{S}_m)$ is a simple matroid of rank 4 on $q^4 + m$ elements, with proper nonempty flats given by

(1) rank 1: the singletons of E;
(2) rank 2: the affine lines and the two-element subsets of E meeting X;
(3) rank 3: the affine planes outside $\bigcup_i \mathscr{A}_i$, the sets $P \cup \{p_i\}$ with $P \in \mathscr{A}_i$, and the three-element subsets of E containing at least two elements of X.

In particular, M_q is a simple matroid of rank 4 on $q^4 + \frac{q^2+1}{2}$ elements.

Proof. We induct on the number of absorptions, starting from $\mathrm{M}_0 = \mathrm{Tr}^1(\mathrm{AG}(4,q))$. Since the spreads are pairwise disjoint, the planes in $\mathscr{A}_i$ remain hyperplanes of M_{i-1}. Since members of $\mathscr{A}_i$ intersect in rank at most 1, Proposition 3.11 implies that the resulting matroid is M_i. Adjoining p_i as a coloop and truncating once retains the old hyperplanes and creates the hyperplanes $L \cup \{p_i\}$ for L a flat of rank 2 of M_{i-1}. Since every affine line lies in exactly one plane of $\mathscr{A}_i$, coalescence replaces each selected plane P by $P \cup \{p_i\}$ and removes precisely the hyperplanes $L \cup \{p_i\}$ with L an affine line. The other new hyperplanes are exactly the listed triples containing p_i. The new flats of ranks 1 and 2 in the intermediate matroid are $\{p_i\}$ and $\{z, p_i\}$ for $z \in E(\mathrm{M}_{i-1})$. Since hyperplane coalescence preserves all flats of these ranks, this establishes the inductive step. □

For an odd prime power q, a *packing*, introduced by [Den72, Den73], is a partition of all lines of $\mathrm{PG}(3,q)$ into $q^2 + q + 1$ spreads.

Definition 4.10 (The affine packing family) For an odd prime power q, let $\{\mathcal{S}_1, \ldots, \mathcal{S}_{q^2+q+1}\}$ be a packing. The *affine packing family* is defined by

$$\mathrm{M}_q^{\mathrm{pk}} \coloneqq \mathrm{M}(\mathcal{S}_1, \ldots, \mathcal{S}_{q^2+q+1}).$$

The matroid $\mathrm{M}_q^{\mathrm{pk}}$ is a simple matroid of rank 4 on $q^4 + q^2 + q + 1$ elements.

4.1.1. *The flag f-vector of an affine family.* We compute the chain and h-polynomials by first counting flats and then incidences between them. Fix pairwise disjoint spreads $\mathcal{S}_1, \ldots, \mathcal{S}_m$ of $\mathrm{PG}(3,q)$, and write $\mathrm{M} = \mathrm{M}(\mathcal{S}_1, \ldots, \mathcal{S}_m)$. Let N_1, N_2, N_3 denote the

numbers of points, affine lines, and affine planes in $V = \mathbb{F}_q^4$, respectively. Then

$$\begin{aligned} N_1 &= q^4, \\ N_2 &= q^3(q^3+q^2+q+1), \\ N_3 &= q^2(q^2+1)(q^2+q+1). \end{aligned}$$

Indeed, there are q^3+q^2+q+1 line directions and $(q^2+1)(q^2+q+1)$ plane directions, with q^3 and q^2 affine translates per direction, respectively. Each spread contains q^2+1 plane directions, so

$$|\mathcal{A}(\mathcal{S}_i)| = q^2(q^2+1).$$

Since the spreads are pairwise disjoint, their selected affine planes are distinct.

Let W_i denote the number of flats of rank i, or *the ith Whitney number* of M. The description of flats of M in Proposition 4.9 implies the following formula for W_i:

$$\begin{aligned} W_1 &= N_1 + m, \\ W_2 &= N_2 + N_1 m + \binom{m}{2}, \\ W_3 &= N_3 + N_1\binom{m}{2} + \binom{m}{3}. \end{aligned}$$

Corollary 4.11 *The flag f-vector of* M *is*

$$\begin{aligned} \alpha(i) &= W_i, \\ \alpha(12) &= qN_2 + 2N_1 m + 2\binom{m}{2}, \\ \alpha(13) &= q^2N_3 + mq^2(q^2+1) + 3N_1\binom{m}{2} + 3\binom{m}{3}, \\ \alpha(23) &= (q^2+q)N_3 + mq^4(q^2+1) + 3N_1\binom{m}{2} + 3\binom{m}{3}, \\ \alpha(123) &= q(q^2+q)N_3 + 2mq^4(q^2+1) + 6N_1\binom{m}{2} + 6\binom{m}{3}. \end{aligned}$$

In particular, the flag f-vector of M *depends only on q and m; the choices $m = (q^2+1)/2$ and $m = q^2+q+1$ give the flag f-vectors of M_q for odd prime powers q and $\mathrm{M}_q^{\mathrm{pk}}$ for all prime powers q, respectively.*

Proof. It is immediate that $\alpha(\varnothing) = 1$, and $\alpha(i) = W_i$. There are N_2 affine lines, each containing q points, and $N_1 m + \binom{m}{2}$ two-point lines containing an auxiliary point. Counting points on these lines gives

$$\alpha(12) = qN_2 + 2\left(N_1 m + \binom{m}{2}\right).$$

For the other entries, we group chains by their largest flat:

(1) There are $N_3 - mq^2(q^2+1)$ unchanged affine planes, each containing q^2 points and q^2+q lines, and contributing q^2, q^2+q, and $q(q^2+q)$ to $\alpha(13)$, $\alpha(23)$, and $\alpha(123)$, respectively.

(2) There are $mq^2(q^2+1)$ enlarged planes $P \cup \{p_i\}$, each containing q^2+1 points and $2q^2+q$ lines: the q^2+q affine lines of P, each of size q, and the q^2 two-point

lines $\{x, p_i\}$ with $x \in P$, and contributing q^2+1, $2q^2+q$, and $q(q^2+q)+2q^2$ to $\alpha(13)$, $\alpha(23)$, and $\alpha(123)$, respectively.

(3) There are $N_1\binom{m}{2} + \binom{m}{3}$ three-element hyperplanes, each containing three points and three lines, all of size 2, and contributing 3, 3, and 6 to $\alpha(13)$, $\alpha(23)$, and $\alpha(123)$, respectively.

Summing these contributions over all hyperplanes gives the stated formulas. □

4.2. **The Gabidulin family.** We first describe absorption in a truncated projective geometry. A Gabidulin code will then provide the families of subspaces needed for this construction.

Fix a prime power q and integers $n \geqslant k \geqslant 3$, and let $V = \mathbb{F}_q^n$. For a linear subspace $X \subseteq V$, write $\mathbb{P}(X)$ for its set of one-dimensional subspaces, with $\mathbb{P}(0) = \varnothing$. The matroid

$$\mathrm{M}_0 = \mathrm{Tr}^{n-k}\big(\mathrm{PG}(n-1, q)\big)$$

has rank k and ground set $\mathbb{P}(V)$. Here Tr^0 is the identity. Its proper flats are the sets $\mathbb{P}(X)$ with $\dim X \leqslant k-1$, of matroid rank $\dim X$.

Let $\mathcal{S}_1, \ldots, \mathcal{S}_m$ be pairwise disjoint families of $(k-1)$-dimensional linear subspaces of V, satisfying

$$\dim(X \cap X') \leqslant k-3 \qquad \text{for distinct } X, X' \in \mathcal{S}_i. \tag{19}$$

The corresponding hyperplanes $\mathbb{P}(X)$ are called *geometric blocks*. We next adjoin one point for each family and absorb it into every geometric block of that family.

Definition 4.12 Let $X_0 = \{p_1, \ldots, p_m\}$ be a set of distinct auxiliary points disjoint from $\mathbb{P}(V)$. Starting from M_0, define M_i for $1 \leqslant i \leqslant m$ as follows. First adjoin p_i as a coloop and truncate once:

$$\widetilde{\mathrm{M}}_i = \mathrm{Tr}^1\big(\mathrm{M}_{i-1} \oplus \mathrm{U}_{\{p_i\},\{p_i\}}\big).$$

Then obtain M_i from $\widetilde{\mathrm{M}}_i$ by coalescing the blocks

$$\mathscr{H}_i = \{\mathbb{P}(X) \cup \{p_i\} : X \in \mathcal{S}_i\}.$$

The resulting matroid is M_m.

Proposition 4.13 *The construction in Definition 4.12 is well-defined. The proper flats of* M_m *are the following, where* $X \subseteq V$ *is a linear subspace and* $T \subseteq X_0$*:*

(L) *The flats* $\mathbb{P}(X) \cup T$ *with* $\dim X + |T| \leqslant k-2$*, of rank* $\dim X + |T|$*.*

(N) *The unselected geometric hyperplanes* $\mathbb{P}(X)$ *with* $\dim X = k-1$ *and* $X \notin \mathcal{S}_1 \cup \cdots \cup \mathcal{S}_m$*.*

(B) *The coalesced hyperplanes* $\mathbb{P}(X) \cup \{p_i\}$ *with* $X \in \mathcal{S}_i$*.*

(U) *The unabsorbed hyperplanes* $\mathbb{P}(X) \cup \{p_i\}$ *with* $\dim X = k-2$ *and* X *contained in no member of* $\mathcal{S}_i$*.*

(H) *The hyperplanes* $\mathbb{P}(X) \cup T$ *with* $|T| \geqslant 2$ *and* $\dim X + |T| = k-1$*.*

Proof. We induct on the number of absorption steps. The assertion holds for M_0. Suppose it holds for M_{i-1}. Since the families are pairwise disjoint, every $\mathbb{P}(X)$ with $X \in \mathcal{S}_i$ remains a hyperplane of M_{i-1}. The intersection bound (19) therefore allows absorption by Proposition 3.11.

The hyperplanes of $\widetilde{\mathrm{M}}_i$ are the same hyperplanes as those of M_{i-1} and the sets $F \cup \{p_i\}$, where F is a rank-$(k-2)$ flat of M_{i-1}. Hyperplane coalescence replaces each selected $\mathbb{P}(X)$ by $\mathbb{P}(X) \cup \{p_i\}$. Among the new hyperplanes, it removes precisely those $\mathbb{P}(Y) \cup \{p_i\}$ for which Y is contained in a member of $\mathcal{S}_i$. Hyperplanes containing an earlier auxiliary point remain, since they cannot lie in a selected block.

Adjoining p_i and truncating results in the lower flats listed in (L), and these are unchanged by coalescence. The remaining and coalesced hyperplanes give (N), (B), (U), and (H), completing the induction. □

We now compute the flag f-vector when $k = 4$. Let N_1, N_2, N_3 denote the numbers of points, lines, and planes in $\mathrm{PG}(n-1, q)$:

$$N_1 = \begin{bmatrix} n \\ 1 \end{bmatrix}_q, \qquad N_2 = \begin{bmatrix} n \\ 2 \end{bmatrix}_q, \qquad N_3 = \begin{bmatrix} n \\ 3 \end{bmatrix}_q,$$

with $\begin{bmatrix} n \\ i \end{bmatrix}_q$ denoting the number of i-dimensional subspaces of $\mathbb{F}_q^n$. Each projective line contains $q+1$ points, and each projective plane contains $[3]_q = q^2 + q + 1$ points and the same number of lines. The pairwise disjoint families $\mathcal{S}_1, \ldots, \mathcal{S}_m$ select a total of

$$b = \sum_{i=1}^{m} |\mathcal{S}_i|$$

geometric planes. Hyperplane coalescence replaces each selected plane $\mathbb{P}(X)$, with $X \in \mathcal{S}_i$, by $\mathbb{P}(X) \cup \{p_i\}$. Thus there are $N_3 - b$ unchanged geometric planes and b coalesced planes. For each auxiliary point p_i, the other possible hyperplanes containing no further auxiliary point are $\ell \cup \{p_i\}$, where ℓ is a projective line. Such a hyperplane is absorbed precisely when ℓ lies in a plane selected by $\mathcal{S}_i$. Each selected plane contains $[3]_q$ lines, and (19) ensures that no line lies in two planes selected by the same family. Therefore, the number of unabsorbed hyperplanes of this form is

$$N_{\mathrm{U}} = mN_2 - b[3]_q.$$

The remaining hyperplanes contain at least two auxiliary points. By Proposition 4.13, they are the triples consisting of one geometric point and two auxiliary points, or three auxiliary points. As a result, the number of such hyperplanes is

$$N_1 \binom{m}{2} + \binom{m}{3}.$$

Let W_j denote the number of rank-j flats of M_m. The description of the flats above and the counts give

$$\begin{aligned} W_1 &= N_1 + m, \\ W_2 &= N_2 + N_1 m + \binom{m}{2}, \\ W_3 &= N_3 + N_{\mathrm{U}} + N_1 \binom{m}{2} + \binom{m}{3}. \end{aligned}$$

Corollary 4.14 *The flag f-vector of* M_m *is*

$$\begin{aligned}
\alpha(i) &= W_i,\\
\alpha(12) &= (q+1)N_2 + 2N_1 m + 2\binom{m}{2},\\
\alpha(13) &= [3]_q N_3 + b + (q+2)N_{\mathrm{U}} + 3N_1\binom{m}{2} + 3\binom{m}{3},\\
\alpha(23) &= [3]_q N_3 + [3]_q b + (q+2)N_{\mathrm{U}} + 3N_1\binom{m}{2} + 3\binom{m}{3},\\
\alpha(123) &= (q+1)[3]_q N_3 + 2[3]_q b + 3(q+1)N_{\mathrm{U}} + 6N_1\binom{m}{2} + 6\binom{m}{3}.
\end{aligned}$$

Proof. There are N_2 geometric lines, each containing $q+1$ points, and $N_1 m + \binom{m}{2}$ two-point lines involving auxiliary points. Counting points on these lines gives

$$\alpha(12) = (q+1)N_2 + 2\left(N_1 m + \binom{m}{2}\right).$$

For the other entries, group chains by their largest flat.

(1) There are $N_3 - b$ unselected geometric planes. Each contains $[3]_q$ points and $[3]_q$ lines, all of size $q+1$. It therefore contributes $[3]_q$, $[3]_q$, and $(q+1)[3]_q$ to $\alpha(13)$, $\alpha(23)$, and $\alpha(123)$, respectively.

(2) There are b coalesced planes. Each contains $[3]_q+1$ points and $2[3]_q$ lines: the $[3]_q$ geometric lines, each of size $q+1$, and the $[3]_q$ two-point lines joining a geometric point to the auxiliary point. It therefore contributes $[3]_q+1$, $2[3]_q$, and $(q+3)[3]_q$ to $\alpha(13)$, $\alpha(23)$, and $\alpha(123)$, respectively.

(3) There are N_{U} unabsorbed hyperplanes with one auxiliary point. Each contains $q+2$ points and $q+2$ lines: one geometric line of size $q+1$, and $q+1$ two-point lines through the auxiliary point. It therefore contributes $q+2$, $q+2$, and $3(q+1)$ to $\alpha(13)$, $\alpha(23)$, and $\alpha(123)$, respectively.

(4) There are $N_1\binom{m}{2} + \binom{m}{3}$ three-element hyperplanes. Each contains three points and three lines, all of size 2. It therefore contributes 3, 3, and 6 to $\alpha(13)$, $\alpha(23)$, and $\alpha(123)$, respectively.

Summing these contributions over all hyperplanes gives the stated formulas. □

4.3. **The projective family.** Let q be a prime power and $4 \leqslant N \leqslant q^3$. We construct a rank-7 matroid $\mathscr{C}(q,N)$ by hyperplane coalescence, then compute its flag numbers and zeta values. Let $\mathrm{P}_d(q) = \mathrm{PG}(d-1,q)$, with $\mathrm{P}_0(q)$ empty. Let

$$U = \mathbb{F}_q^3, \qquad K = \mathbb{F}_{q^3}, \qquad V = U \oplus K.$$

Let X be a disjoint copy of $\mathbb{Z}/N\mathbb{Z}$, and put

$$\mathrm{Q} = \mathrm{P}_6(q) \oplus \mathrm{U}_{X,X}, \qquad \mathrm{M} = \mathrm{Tr}^{N-1}(\mathrm{Q}).$$

Thus M has rank 7. Its proper flats are $\mathbb{P}(W) \sqcup D$, where $W \subseteq V$ is a linear subspace, $D \subseteq X$, and $\dim W + |D| \leqslant 6$.

Choose an $\mathbb{F}_q$-basis $\alpha_1, \alpha_2, \alpha_3$ of K. For the standard basis e_1, e_2, e_3 of U, define

$$\phi : \mathrm{Hom}_{\mathbb{F}_q}(U, K) \longrightarrow K, \qquad A \longmapsto \sum_{i=1}^{3} \alpha_i A(e_i).$$

Each map A determines a three-dimensional graph

$$G_A = \{(u, A(u)) : u \in U\}.$$

The map ϕ is surjective, so each fiber has q^6 elements. Moreover, its kernel contains no rank-one map: if $A(e_i) = c_i v$, with $v \neq 0$ and $c_i \in \mathbb{F}_q$, then

$$\phi(A) = 0 \quad \Longrightarrow \quad \sum_i c_i \alpha_i = 0 \quad \Longrightarrow \quad A = 0.$$

Consequently, distinct maps in the same fiber satisfy

$$\dim(G_A \cap G_{A'}) = 3 - \mathrm{rk}(A - A') \leqslant 1.$$

Definition 4.15 Fix an injection $\iota : X \to K$. For each four-element subset $B \subseteq X$, let $\sigma(B) = \sum_{b \in B} b$, computed in $\mathbb{Z}/N\mathbb{Z}$. Define

$$\mathscr{H} = \{\mathbb{P}(G_A) \sqcup B : |B| = 4,\ \phi(A) = \iota(\sigma(B))\}.$$

With these choices fixed, the *projective family* $\mathscr{C}(q, N)$ is obtained by coalescing $\mathscr{H}$ in M. There are

$$c := |\mathscr{H}| = q^6 \binom{N}{4}$$

selected blocks.

To verify that coalescence is possible, each block is a proper rank-7 flat of Q, hence a spanning subset of M. Distinct blocks meet in rank at most 5: equal graphs force equal subset sums, so their distinct auxiliary sets share at most two elements; equal auxiliary sets give graph intersections of dimension at most one; otherwise the two contributions are at most two and three. For an independent six-element subset I of a block H,

$$\mathrm{cl}_{\mathrm{M}}(I) = \mathrm{cl}_{\mathrm{Q}}(I) \subseteq H.$$

These are precisely the two conditions of Lemma 3.7.

The resulting matroid is simple of rank 7. If $N \geqslant 13$, it is 3-connected: every bipartition has a part containing seven auxiliary points, which form a retained basis. That part spans, so the connectivity value is the rank of the other part, at least two whenever both parts have at least two elements.

We now count flags. Every selected block has the same old restriction

$$\mathrm{L} = \mathrm{P}_3(q) \oplus \mathrm{U}_{4,4}.$$

Inside it, the old hyperplanes comprise $[3]_q = q^2 + q + 1$ copies of $\mathrm{P}_2(q) \oplus \mathrm{U}_{4,4}$ and four copies of $\mathrm{P}_3(q) \oplus \mathrm{U}_{3,3}$. Coalescence replaces these $[3]_q + 4$ hyperplanes by one and preserves all flats of rank at most 5. Thus the Whitney numbers are

$$W_s = \sum_{a+b=s} \begin{bmatrix} 6 \\ a \end{bmatrix}_q \binom{N}{b} - \begin{cases} 0, & 1 \leqslant s \leqslant 5, \\ c([3]_q + 3), & s = 6. \end{cases}$$

Here $\begin{bmatrix} d \\ a \end{bmatrix}_q$ counts a-dimensional subspaces of $\mathbb{F}_q^d$, and the sum ranges over $0 \leqslant a \leqslant 6$ and $0 \leqslant b \leqslant N$. For a rank set $S \subseteq \{1, \ldots, d + n - 1\}$, let $\alpha_{d,n}(S)$ count flags of proper flats in $\mathrm{P}_d(q) \oplus \mathrm{U}_{n,n}$, with $\alpha_{d,n}(\varnothing) = 1$. Counting by the largest flat gives the recurrence

$$\alpha_{d,n}(S) = \sum_{\substack{0 \leqslant a \leqslant d,\ 0 \leqslant b \leqslant n \\ a+b=s}} \begin{bmatrix} d \\ a \end{bmatrix}_q \binom{n}{b} \alpha_{a,b}(S \setminus \{s\}), \qquad s = \max S.$$

Proposition 4.16 *For $T \subseteq \{1,\dots,5\}$, the flag numbers of $\mathscr{C}(q,N)$ satisfy*

$$\begin{aligned}\alpha_{\mathscr{C}(q,N)}(T) &= \alpha_{6,N}(T),\\ \alpha_{\mathscr{C}(q,N)}(T\cup\{6\}) &= \alpha_{6,N}(T\cup\{6\})\\ &\quad + c\big(\alpha_{3,4}(T) - \alpha_{3,4}(T\cup\{6\})\big).\end{aligned}$$

Proof. Within each selected block, coalescence removes $\alpha_{3,4}(T\cup\{6\})$ flags ending at old hyperplanes and introduces $\alpha_{3,4}(T)$ flags ending at the new hyperplane. Distinct blocks contain no common old hyperplane, and these corrections are additive, so we obtain the desired count. □

4.4. **Multichain counts are not log-concave.**

Proof of Theorem B. For each family $\mathrm{M} = \mathrm{M}(q)$ of rank r, define for $j = 1,2,3$,

$$c_j = \sum_{\substack{S\subseteq[r-1]\\ |S|=j}} \alpha_{\mathrm{M}}(S), \qquad D(q) = Z_{\mathrm{M}}(3)^2 - Z_{\mathrm{M}}(2)Z_{\mathrm{M}}(4).$$

By (3), these counts yield multichain counts

$$\begin{aligned}Z_{\mathrm{M}}(2) &= 2 + c_1,\\ Z_{\mathrm{M}}(3) &= 3 + 3c_1 + c_2,\\ Z_{\mathrm{M}}(4) &= 4 + 6c_1 + 4c_2 + c_3.\end{aligned} \tag{20}$$

Then the defect $D(q)$ can be viewed as a polynomial in q. We first determine its sign for $x \in \mathbb{R}_{>0}$. In each family, the leading coefficient of $D(q)$ is negative and all its other nonzero coefficients are positive. This implies that there is exactly one positive zero ξ, so $D(\xi) = 0$ and

$$D(x) > 0 \quad (0 < x < \xi), \qquad D(x) < 0 \quad (x > \xi).$$

The matroids are defined only at the allowed prime powers q. Thus the threshold is the first allowed prime power greater than ξ. We locate this zero between consecutive allowed prime powers by exact evaluation, as follows.

(a) For M_q, use Corollary 4.11 with $m = (q^2+1)/2$. The consecutive odd prime powers 89 and 97 satisfy

$$D(89) > 0 > D(97), \qquad 89 < \xi < 97.$$

Hence, for odd prime powers q, $D(q) < 0$ if and only if $q \geqslant 97$.

(b) For $\mathrm{M}_q^{\mathrm{pk}}$, use Corollary 4.11 with $m = q^2+q+1$. The consecutive prime powers 79 and $81 = 3^4$ satisfy

$$D(79) > 0 > D(81), \qquad 79 < \xi < 81.$$

Hence, for prime powers q, $D(q) < 0$ if and only if $q \geqslant 81$.

(c) For $\mathrm{G}_{4,q}$, use the counts following Proposition 4.13 with $n = 6$, $m = q^3$, and $b = q^9$. Again, the consecutive prime powers 79 and 81 satisfy

$$D(79) > 0 > D(81), \qquad 79 < \xi < 81.$$

Hence, for prime powers q, $D(q) < 0$ if and only if $q \geqslant 81$.

(d) For $\mathrm{C}(q, q^3)$, count chains by their largest proper flat. The ambient truncation has $\begin{bmatrix}6\\a\end{bmatrix}_q\binom{q^3}{b}$ flats of type (a, b), with $1 \leqslant a+b \leqslant 6$, each restricting to $\mathrm{P}_a(q)\oplus\mathrm{U}_{b,b}$. Repeat this count inside each restriction, then apply Lemma 3.9 to the $q^6\binom{q^3}{4}$ blocks. The consecutive prime powers 139 and 149 satisfy

$$D(139) > 0 > D(149), \qquad 139 < \xi < 149.$$

Hence, for prime powers q, $D(q) < 0$ if and only if $q \geqslant 149$.

Since $D(q) = Z_{\mathrm{M}}(3)^2 - Z_{\mathrm{M}}(2)Z_{\mathrm{M}}(4)$, these are precisely the four equivalences in the theorem. Its final assertion follows from Theorem A. □

4.5. **Three-connected rank-7 counterexamples.**

Proof of Theorem C. Let $\mathrm{X} = \mathrm{M}_q$ for an odd prime power q, or $\mathrm{X} = \mathrm{G}_{4,q}$ for any prime power q, and put

$$\mathrm{Y} = \mathrm{Tr}^2(\mathrm{X} \oplus \mathrm{X} \oplus \mathrm{U}_{1,1}).$$

The direct sum is simple of rank 9 and hence its truncation is connected and simple, and its rank-7 truncation is simple by Lemma 3.2.

We count flags in $\mathscr{L}(\mathrm{X})^2 \times \mathscr{L}(\mathrm{U}_{1,1})$ by their coordinate ranks, retaining total rank sets in [6]. For each choice of weakly increasing coordinate ranks, the chains in the factors are chosen independently. Substitution in Eqs. (10) and (11) gives the Chow coefficients $\underline{\mathrm{H}}_i(q)$ and the cubic γ-polynomial Γ_q. Let

$$R(q) = \mathrm{disc}(\Gamma_q), \qquad \Delta(q) = \underline{\mathrm{H}}_2(q)^2 - \underline{\mathrm{H}}_1(q)\underline{\mathrm{H}}_3(q).$$

The Taylor coefficients of $\gamma_3(q)$ at $q_{\min} = 3$ for the affine family, and at $q_{\min} = 2$ for the Gabidulin family, are nonnegative, with positive constant term. Hence $\gamma_3(q) > 0$ for $q \geqslant q_{\min}$, and Lemma 2.9 implies that

$$\begin{aligned} \underline{\mathrm{H}}(\mathrm{Y}; t) \text{ is not real-rooted} &\iff R(q) < 0, \\ \underline{\mathrm{H}}(\mathrm{Y}; t) \text{ is not log-concave} &\iff \Delta(q) < 0. \end{aligned}$$

We first determine the signs of R and Δ for real arguments $x \geqslant q_{\min}$. For either $D = R$ or $D = \Delta$, the polynomial $D(q_{\min} + x)$ has one sign variation, a positive constant term, and a negative leading coefficient. Descartes' rule therefore implies that $D(q)$ has exactly one zero $\xi_D > q_{\min}$, with

$$D(\xi_D) = 0, \qquad D(x) > 0 \quad (q_{\min} \leqslant x < \xi_D), \qquad D(x) < 0 \quad (x > \xi_D).$$

Thus ξ_R and ξ_Δ indicate where the discriminant and the log-concavity defect, respectively, change sign as q varies. We locate these zeros between consecutive allowed prime powers by computer computations.

(a) For $\mathrm{X} = \mathrm{M}_q$, use Corollary 4.11 with $m = (q^2+1)/2$. The consecutive odd prime powers $43, 47$ and $89, 97$ satisfy

$$\begin{aligned} R(43) > 0 > R(47), &\quad 43 < \xi_R < 47, \\ \Delta(89) > 0 > \Delta(97), &\quad 89 < \xi_\Delta < 97. \end{aligned}$$

Hence, for odd prime powers q, $R(q) < 0$ if and only if $q \geqslant 47$, and $\Delta(q) < 0$ if and only if $q \geqslant 97$.

(b) For $X = G_{4,q}$, use the counts following Proposition 4.13 with $n = 6$, $m = q^3$, and $b = q^9$. The consecutive prime powers $32 = 2^5, 37$ and $79, 81 = 3^4$ satisfy

$$R(32) > 0 > R(37), \quad 32 < \xi_R < 37,$$
$$\Delta(79) > 0 > \Delta(81), \quad 79 < \xi_\Delta < 81.$$

Hence, for prime powers q, $R(q) < 0$ if and only if $q \geqslant 37$, and $\Delta(q) < 0$ if and only if $q \geqslant 81$.

By the two criteria above, these are the desired real-rootedness and log-concavity thresholds. □

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